



From rule-based models to deep learning transformers architectures for natural language processing and sign language translation systems: survey, taxonomy and performance evaluation

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Abstract

With the growing Deaf and Hard of Hearing population worldwide and the persistent shortage of certified sign language interpreters, there is a pressing need for an efficient, signs-driven, integrated end-to-end translation system, from sign to gloss to text and vice-versa. There has been a wealth of research on machine translations and related reviews. However, there are few works on sign language machine translation considering the particularity of the language being continuous and dynamic. This paper aims to address this void, providing a retrospective analysis of the temporal evolution of sign language machine translation algorithms and a taxonomy of the Transformers architectures, the most used approach in language translation. We also present the requirements of a real-time Quality-of-Service sign language machine translation system underpinned by accurate deep learning algorithms. We propose future research directions for sign language translation systems.

Keywords Artificial intelligence · Deep learning · Natural language processing · Neural machine translation · Sign language translation, · Transformers

Abbreviations

AR	Augmented Reality
ArSL	Arabic Sign Language
ASL	American Sign Language
CNN	Convolution Neural Networks
CSLR	Continuous Sign Language Recognition
CTC	Connectionist Temporal Classification
DHH	Deaf or Hard of Hearing
DNN	Deep Neural Network
EBMT	Example-Based MT
FFT	Fast Furrier Transform

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FKNN	Fuzzy-KNN
G2S	Gloss-to-Sign
G2T	Gloss-to-Text
GNN	Graph Neural Networks
GPT	Generative Pre-trained Transformers
GRU	Gated Recurrent Unit
IoT	Internet-of-Things
ISLR	Isolated Sign Language Recognition
KNN	K-Nearest Neighbors
LSTM	Long-Short Term Memory
MT	Machine Translation
NLP	Natural Language Processing
NMT	Neural MT
POS	Part-of-Speech
QoS	Quality-of-Service
RBMT	Rule-Based MT
S2G	Sign-to-Gloss
S2G2T	Sign-to-Gloss-to-Text
S2T	Sign-to-Text
SLMT	Sign Language Machine Translation
SLT	Sign Language Translation
SMT	Statistical MT
T2G	Text-to-Gloss
VR	Virtual Reality

1 Introduction

A Deaf or Hard of Hearing (DHH) person is someone who has hearing loss. According to the World Health Organization, there are approximately 430 million DHH worldwide, which is estimated to double by 2050 (World Health Organization 2023). The means of communication for such a person is sign language, a visual language that relies on the movement of different body parts and facial expressions to convey meaning instead of spoken words (Sandler and Lillo-Martin 2006). Noting that there are 300 sign languages globally (Kumar et al. 2015) that are independent of spoken languages and have their grammar and syntax (Jay 2011). To establish communication between the DHH and the broader society, sign language interpreters act as intermediaries to interpret between sign and spoken languages. However, there is a shortage of interpreters. For instance, there are around 10,000 certified interpreters in the United States (Registry of Interpreters for the Deaf 2023), while there are about 48 million Deaf individuals residing in the country (Healthy Hearing 2023). This scarcity of interpreters, combined with the increasing number of the DHH population, actuates the introduction of real-time automated sign language translation systems. In particular, in the era of smart cities, the population's well-being is essential (Ismail and Zhang 2018; Ismail et al. 2020), developing real-time smart translation systems for the DHH is necessary to provide safer, healthier, and more pleasant experiences. The ability to enable communication between DHH and the hearing population could be life-saving in a disastrous event such as a medical emergency (Tonkin 2017). Consequently,

building a software-based real-time translation system that accurately and efficiently translates sign languages to spoken languages and vice versa is crucial.

Machine Translation (MT) was introduced as part of Natural Language Processing (NLP) to translate one spoken language to another (Stahlberg 2020). MT algorithms evolved from rule-based to neural networks and are classified into four categories (Poibeau 2017): Rule-Based MT (RBMT), Example-Based MT (EBMT), Statistical MT (SMT), and Neural MT (NMT). Several works exist on machine translation for spoken languages (Johnson et al. 2017; Zhang et al. 2023). Despite that MT evolution faced a lot of challenges, including language and context complexity (Arivazhagan et al. 2019), idiomatic expressions (Dankers et al. 2022), and time efficiency (Stahlberg 2020), there have been a lot of advances in the development of efficient spoken end-to-end translation systems (Stahlberg 2020). On the other hand, regarding sign language translation, several studies have applied MT techniques to convert sign language into spoken language and vice versa. These efforts typically concentrate on translating discrete components, either from sign to gloss (Camgoz et al. 2020), where signs are the image/video frames and gloss are the linguistic representations of signs, or from gloss to text (Camgoz et al. 2018). The introduction of gloss contributes to higher precision when translating from sign to text (Li and Meng 2022), or when generating signs (Huang et al. 2022). In addition, several existing surveys explored sign language translations in the literature (Papastratis et al. 2021; Núñez-Marcos et al. 2023; Farooq et al. 2021; Kahlon and Singh 2023; El-Alfy and Luqman 2022). However, they tackled particular aspects of the sign language translations, such as direct translation from signs to spoken language text (Papastratis et al. 2021; Núñez-Marcos et al. 2023; El-Alfy and Luqman 2022), signs to gloss (Papastratis et al. 2021; Farooq et al. 2021; Kahlon and Singh 2023; El-Alfy and Luqman 2022), gloss to spoken language text (Papastratis et al. 2021; Núñez-Marcos et al. 2023; Farooq et al. 2021; Kahlon and Singh 2023), spoken language text to gloss (Papastratis et al. 2021; Núñez-Marcos et al. 2023; Farooq et al. 2021), or gloss to signs (Papastratis et al. 2021; Núñez-Marcos et al. 2023; Farooq et al. 2021; Kahlon and Singh 2023). In our survey, we present a holistic approach where sign language translation enables communication between DHH and hearing individuals in a seamless end-to-end framework from sign to gloss to text and backward, presented in terms of stages. Furthermore, we provide an in-depth analysis of the evolution and current state of the MT systems, particularly focusing on sign language translation revealing insights, challenges, and future research directions. Furthermore, we present a Transformer-based translation case study comparing the performances of its different architectures via empirical evaluations.

In this work, we define a Sign Language Machine Translation (SLMT) algorithm as the composition of two parts that we designate as: (1) Sign-to-Gloss (S2G) recognition, and (2) Sign Language Translation (SLT). S2G falls under computer vision and includes converting static and dynamic gestures and movements in sign language video frames into a corresponding form. This part does not consider the grammar and context in the recognition process. On the other hand, SLT involves translating the recognized content from sign language into spoken language and vice versa while considering both languages' grammar and syntax. This part of the system can be divided into the following subparts: Gloss-to-Text (G2T) and Text-to-Gloss (T2G). The former translates the spoken language text into sign language gloss, while the latter reverses this process.

In this paper, we investigate the SLMT algorithms. We explore all fundamental translation directions by providing a comprehensive analysis of the state-of-the-art, offering insights into the research in this domain. We aim to provide a valuable resource for

researchers and developers building deep learning-based end-to-end sign language translation systems. Our main contributions are as follows:

- We present a taxonomy for the sign language components in terms of detection types, sign categories, grammar components, and their interactions.
- We propose a conversational end-to-end SLMT framework that includes the stages involved in the translation process, providing a systematic approach for SLMT researchers and developers to examine translation technologies.
- We compare the SLMT public datasets in terms of size, resolution, and available features such as video, text, and gloss.
- We provide a retrospective analysis of the temporal evolution of SLMT architectures, intending to uncover the challenges that led to the development of the underlying algorithms.
- We classify the SLMT transformer architectures based on the temporal evolution of its functionalities.
- We conduct empirical evaluations of the different transformer architectures in a G2T sign language translation scenario.
- We create a medical-related ASL dataset to perform our empirical evaluations of G2T sign language translation.

The rest of the paper is organized as follows. Section 2 studies the related surveys for sign language translation. Section 3 provides a background on sign language and a taxonomy. Section 4 discusses the machine translation evolution. Section 5 presents the sign language translation framework. Section 6 provides the public sign language datasets. Section 7 discusses the performance evaluation metrics. SLMT algorithms and taxonomy are presented in Sect. 8. Section 9 presents the taxonomy of transformer architecture. A case study of translating gloss to text is presented in Sect. 10. Section 11 presents the challenges and proposed solutions. Section 12 proposes future research directions. Lastly, Sect. 13 concludes the paper.

2 Related surveys for sign language translation

Several reviews on SLMT exist in the literature (Papastratis et al. 2021; Núñez-Marcos et al. 2023; Farooq et al. 2021; Kahlon and Singh 2023; El-Alfy and Luqman 2022). (Papastratis et al. 2021), covering the period from 2018 to 2021 explores the advantages, limitations, and challenges of the different methods. It also analyzes these methods based on the performance achieved in the literature. Moreover, it presents sign language datasets, covering both isolated (I) and continuous (C) sign language translation. However, it does not delve into machine translation evolution or provide taxonomies for sign language applied algorithms or transformer architectures. Nevertheless, this survey includes discussions on T2G, gloss-to-sign (G2S), S2G, and G2T translations, but omits sign-to-gloss-to-text (S2G2T) and sign-to-text (S2T) translations. (Núñez-Marcos et al. 2023) spans a broader period from 2016 to 2022, providing analysis of machine translation evolution, sign language datasets, and continuous sign language data. It also explores the different deep-learning algorithms applied in sign language translation while omitting S2G and S2G2T translations. Nevertheless, it provides partial performance analysis for the sign language translation works, focusing on those post-2018. Despite its comprehensive analysis,

this study lacks providing any taxonomies. (Farooq et al. 2021) examines works from 2015 to 2020, exploring sign language datasets (both isolated and continuous) and presenting taxonomies for sign language and the applied algorithms. However, it does not cover machine translation evolution or transformer architectures. It includes discussions and performance analysis for T2G, G2S, S2G, and G2T translations but does not address S2G2T or S2T translations. (Kahlon and Singh 2023) provides a broader view of the literature from the 1990s to 2020, addressing machine translation evolution, sign language datasets, and performance analysis of the isolated and continuous sign language translation, including T2G, G2S, S2G, and G2T translations. However, it does not cover S2G2T and S2T translations, nor provide any related taxonomies. (El-Alfy and Luqman 2022) presents a comprehensive survey from the 1990s to 2021, exploring sign language acquisition, recognition, translation, and linguistic structures, providing a sign language taxonomy. However, it does not delve into the algorithmic approaches applied in the literature nor provides a related taxonomy. Nevertheless, despite providing performance analysis for the literature, its translation coverage is limited, addressing only G2S and S2T translations.

On the other hand, our survey, spanning from 2016 to 2023, represents a unique and comprehensive effort in exploring SLMT. We delve into critical aspects such as MT evolution and sign language datasets for isolated and continuous signs. We investigate both machine-learning and deep-learning algorithms and provide a thorough performance evaluation. What sets our work apart is our discussion of the architectural components in the literature and our presentation of detailed taxonomies for both sign language and the algorithms applied in SLMT. Importantly, we cover all aspects of the translation process, from T2G to G2T, including intermediate and combined processes like S2G, G2S, and the full S2T translation, supported by an in-depth exploration and taxonomy of transformer architectures. In addition, unlike previous surveys, our survey provides empirical evaluations of the different transformer architectures in a G2T sign language translation scenario. This comprehensive approach underscores our unique contribution to the field, offering insights and classifications not previously provided, as detailed in Table 1.

In summary, while existing surveys on SLMT provide valuable insights, they each have limitations in scope and coverage. Our survey addresses these gaps by comprehensively analyzing machine translation evolution, datasets, algorithms, performance metrics, and transformer architectures. We provide detailed taxonomies covering all translation processes and conduct empirical evaluations of the transformer architectures for G2T translation using public and private datasets, making our work a significant contribution to the field of SLMT.

3 Sign language taxonomy

Sign language conveys meaning visually and has distinct linguistic properties that differ from spoken languages (Jay 2011). Figure 1 presents our taxonomy of sign language. We classify the signs based on their sign detection type, sign category, and grammar composition.

- **Sign Detection.** This category includes the techniques that are used to recognize and interpret sign language. It divides the recognition process into static detection,

Table 1 Comparison between sign language translation-related surveys

Work	Period covered	Domains of Comparison				Covered Translation							
		Machine Translation Evolution	Datasets	Isolated (I)/ Continuous (C) / Both (B)	Performance	Sign Language Taxonomy	Algorithms Taxonomy	Transformer Architectures Taxonomy	Empirical Evaluation	Text-to-Gloss	Gloss-to-Sign	Sign-to-Gloss	Gloss-to-Text
Papastratis et al. (2021)	2018–2021	✓	✓	B	✗	✗	✗	✗	✗	✓	✓	✓	✗
Núñez-Marcos et al. (2023)	2016–2021	✓	✓	C	✗	✗	✗	✗	✗	✓	✓	✗	✓
Farooq et al. (2021)	2015–2020	✗	✓	B	✓	✓	✓	✗	✗	✓	✓	✗	✗
Kahlon and Singh (2023)	1990s–2020	✓	✗	B	✓	✗	✗	✗	✗	✓	✓	✗	✗
El-Alfy and Luqman (2022)	1990s–2021	✗	✓	B	✓	✓	✗	✗	✗	✗	✗	✓	✓
This Work	2016–2023	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓

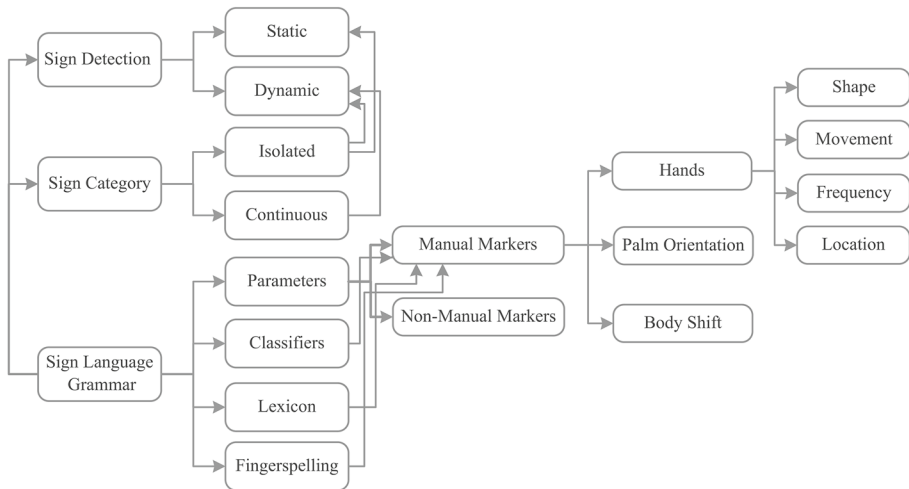


Fig. 1 Sign Language taxonomy

representing static signs, and dynamic detection, reflecting the multiple frames or videos in which sign language is produced.

- **Sign Category.** We classify signs as isolated recognition, representing words, and continuous recognition, representing sentences. This classification recognizes the varying complexities of sign language communication and how it can adapt to scenarios.
- **Sign Language Grammar.** We categorize the elements that shape the structure and meaning of sign language based on its phonology. Phonology, in any language, is a subfield of linguistics that deals with the study of patterns and phonemes (Lass 1984). The phonological parameters of a sign language are categorized into manual and non-manual markers (Jay 2011). The manual markers include (a) handshape, which refers to the specific shape of the fingers and hands while producing signs. (b) handshape movement and frequency, which involves the direction and way the hands and arms move to depict different words and concepts. (c) handshape location, where the signs are produced in relation to the body. (d) palm orientation refers to the position and direction of the palms when producing a sign. (e) body shift, which conveys meaningful messages during conversations and storytelling. Figure 2 shows an example of these markers. On the other hand, the non-manual markers include body language, eye gaze, and facial expressions such as eyebrows and lip movement. This phonology is crucial in conveying emotions, intensity, and grammatical information. In addition, sign language classifiers are linguistic representations of more than words, allowing the signer to provide more detailed descriptions of objects, people, or events. Figure 3 shows two different classifiers. Signers utilize classifiers to add depth, context, and specificity to their conversations, making sign language expressive. Moreover, the sign language lexicon refers to the entire set of signs or lexical items to convey meaning. The last grammar element is fingerspelling, which involves manual signs representing the letters of spoken language. It is commonly used to spell words or names for which no established sign exists, such as proper nouns, technical terms, or foreign words (Jay 2011).

Fig. 2 The sign “Drive” in American Sign Language (Vicars 2023a). Closed fists represent the handshape. The movement mimics the motion of turning a steering wheel. The frequency is twice. The handshape location is in front of the body, near the chest. The palm orientation is inwards

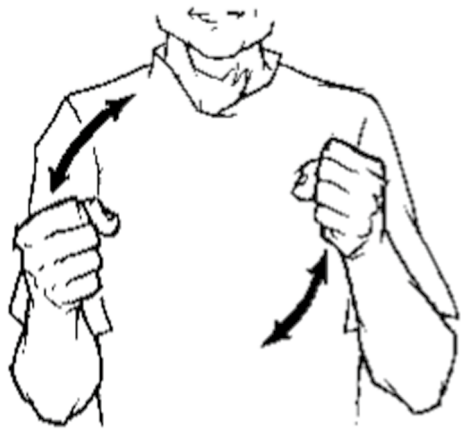


Fig. 3 Examples of classifiers. The gesture on the left represents long, thin things, such as people, while the right represents vehicles such as cars (Vicars 2023b)

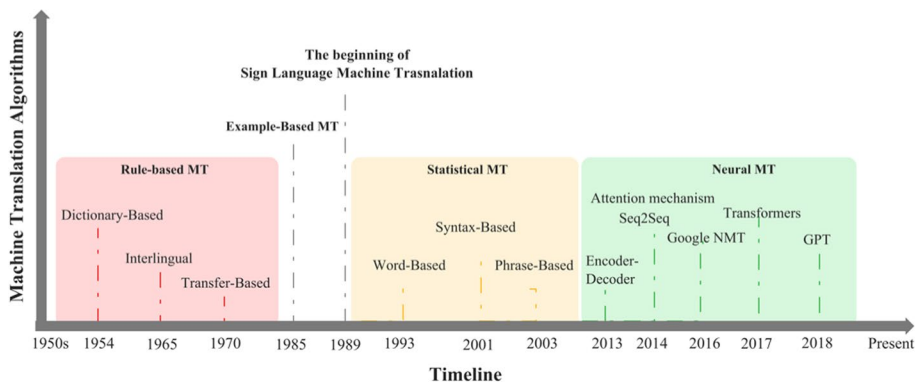
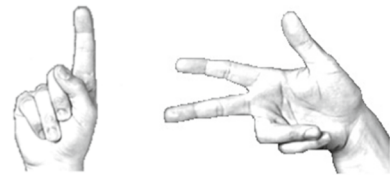


Fig. 4 Machine translation algorithms evolution over time

4 Machine translation evolution

Figure 4 shows the evolution of MT algorithms over time divided into four categories: Rule-Based Machine Translation (RBMT), Example-Based Machine Translation (EBMT), Statistical Machine Translation (SMT), and Neural Machine Translation (NMT) (Wang et al. 2022).

Machine Translation (MT) began in the mid-twentieth century due to the world's communication needs. The first documented effort in MT was developing the Automatic Language Processing Advisory Committee project in the United States in 1966 (National

Research Council 1966.doi: [10.17226, 9547](https://doi.org/10.17226/9547) 1966). The early approach relied on linguistic and grammatical rules to translate languages through a rule-based model (Kay 1984). However, the model was unsatisfactory due to the languages' complexity. The evolution of MT initially focused on spoken languages. However, it extended to sign languages in the 1980s (Núñez-Marcos et al. 2023; Kamata et al. 1989), addressing this visual communication's unique challenges and linguistic nuances. In the following decades, MT research progressed to include statistical and neural approaches in the 1990s and 2000s, respectively, improving translation accuracy (National Research Council 1966.doi: [10.17226, 9547](https://doi.org/10.17226/9547) 1966; Lopez Aug. 2008).

RBMT, introduced in 1954, relies on rules and dictionaries to convert text from one language to another (Kay 1984). It consists of three models: dictionary-based, interlingual, and transfer-based (Nirenburg 1989). Dictionary-based translation combines dictionaries and grammatical rules to translate text from one language to another by breaking the sentence in the source language into smaller units, such as words or short phrases. The model then searches each unit in a bilingual dictionary to map words or phrases in the source language to their corresponding translations in the target language while applying grammatical rules to maintain proper syntax. Although this approach is straightforward, it falls short in handling the language context (Nirenburg 1989).

Interlingual MT was introduced to solve the accuracy problem of the dictionary-based MT. This approach identifies the structure and semantics of the text in the source language. Then, it transforms the linguistic meaning into an intermediate language-neutral globally unified representation. The intermediate representation is then used to generate the target language text. This approach raised concerns regarding its efficiency. Therefore, researchers proposed the Transfer-based approach, which functions like the interlingual translation. However, the intermediate representation in the new approach emphasizes structural transfer rather than meaning transfer and is not unified across languages (Nirenburg 1989).

In contrast to RBMT, EBMT was introduced in the mid-1980s to solve the efficiency and accuracy problems of the RBMT (Wang et al. 2022). This was done by utilizing a database of translated sentences for translation reference. The model identifies and modifies examples from the database to generate translations, making this approach effective for handling idiomatic expressions (Carl and Way 2003).

Furthermore, SMT, introduced in the early 1990s, utilizes probabilistic models and parallel corpora rather than relying on RBMT to produce language translations (Koehn 2009). The fundamental work by IBM laid the foundational models for SMT, introducing key concepts such as the noisy channel model for learning translation probabilities (Brown et al. 1993). Over the years, SMT evolved through several phases, including the introduction of phrase-based models (Zens et al. 2002), which consider sequences of words or phrases instead of word-for-word translations. Nevertheless, SMT faced challenges in addressing the syntactic differences between languages despite its significant improvements in automatic translation systems, including the ability to rapidly adapt to new languages and domains. These challenges highlighted the need for data-driven approaches that were built based on the groundwork of SMT (Wang et al. 2022).

More recently, NMT was introduced in 2016 to represent the recent advancement in machine translation (Wolk and Marasek 2015). NMT models primarily include encoder-decoder (Neco and Forcada 1997), sequence to sequence (Seq2Seq) (Sutskever et al. 2014), and attention mechanisms (Bahdanau et al. 1409). These models analyze the source text and generate target language output incrementally while calculating translation probabilities to determine the best translation output for each word or phrase. Nevertheless, more recently, NMT expanded to include learning models such as Google NMT (Arivazhagan

et al. 1907) and Generative Pre-trained Transformers (GPT) for language translation (Radford et al. 2018). It is worth noting that GPT has four different versions. GPT-1 (Radford et al. 2018), released in 2018, was trained on 5 GB of books. Although this model could generate, summarize, complete, and translate text, it suffered from several limitations, such as limited domain-specific knowledge and weak generalization. GPT-2 (Radford et al. 2019), released in 2019, was trained on 40 GB of Reddit links. This version differs from the previous one by acting as a conversational agent and having a better generalization. Moreover, GPT-3 (Brown et al. 2020), released in 2020, was trained on a much larger dataset (753 GB) based on Wikipedia, books, academic journals, Reddit links, and common crawl. Lastly, GPT-4 (OpenAI 2023), released in 2023, was trained on a 20 TB dataset. Unlike previous versions, which take text only as input, GPT-4 accepts images as input. However, it still outputs text only. In addition, although all versions were trained on data before 2021, GPT-4 can browse the internet through plugins. Recently, (Shahin and Ismail 2023) conducted experiments using ChatGPT to give insights into its potential for G2T and T2G translations. ChatGPT has promising capabilities in translating from spoken English and Arabic languages to Arabic (ArSL) American (ASL), Australian (AUSLAN), and British (BSL) sign languages and vice-versa.

5 Proposed conversational end-to-end sign language machine translation framework

SLMT is a process that involves seamless conversion between sign language and spoken language in both directions, supported by the computer vision and NLP domains. Computer vision involves processing visual data (Parker 2010), which helps understand and generate signs. NLP (Lane et al. 2018), on the other hand, contributes to translating these expressions into spoken language. In this section, we present a framework for SLMT in terms of stages to describe the data flow used in the translation process. Figure 5 illustrates the stages of our framework.

- **Stage 1: Data Collection of Gestures.** Various devices can capture the sign input, including sensor-based wearable devices (Li et al. 2022), such as gloves and armbands, and cameras. These cameras can either be vision-based (Camgoz et al. 2018) or vision-sensor-based, such as Microsoft Kinect (Al-Hammadi et al. 2020; Amin et al. 2022). These instruments can acquire different sign language parameters, such as physical gestures produced by the hands, body movement, and facial expressions, to convey rich linguistic information.
- **Stage 2: Computer Vision for Gesture Recognition.** The signs collected from the previous stage are transformed into gloss, facilitating the S2G process. This builds a connection between gestural communication and linguistic representation, a fundamental element for an accurate translation (Zhou et al. 2021). Computer vision algorithms can perform this transformation by incorporating data augmentation to enhance the dataset, pre-processing to clean and prepare the data for further analysis, and feature extraction to acquire the glosses for translation (Stahlberg 2020).
- **Stage 3: Translation to Text.** NLP models translate sign language gloss to spoken language text through G2T translation. This is achieved by applying tokenization and feature extraction methods to comprehend the context and linguistic patterns (Chen et al.

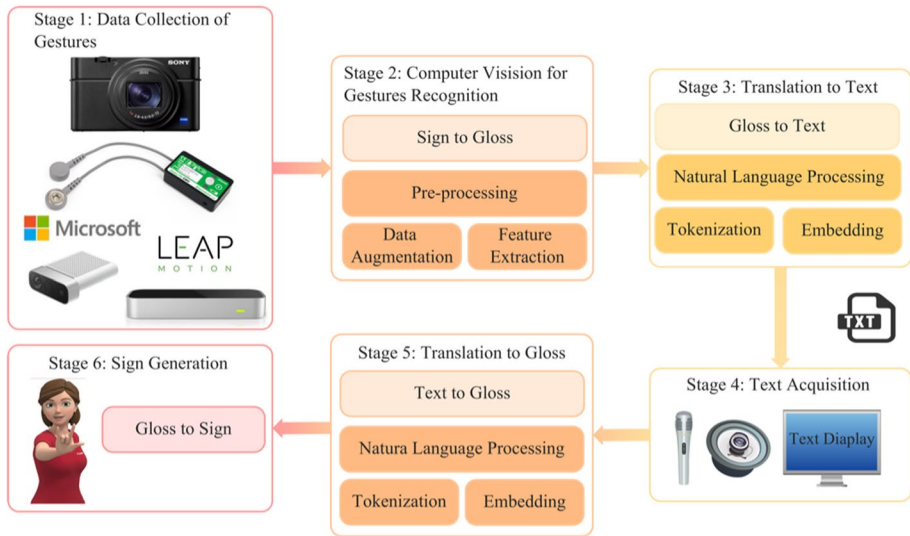


Fig. 5 Sign Language Translation Framework

2022). Individuals who do not know sign language will be able to understand the output of this translation.

- **Stage 4: Text Acquisition.** Different devices serve as sources for spoken language text input including speech-to-text conversion systems, microphones, or other audio input devices.
- **Stage 5: Translation to Gloss.** This is the reverse direction of the G2T translation, which converts written spoken language text into gloss through T2G via similar steps (Kouremenos et al. 2018).
- **Stage 6: Sign Generation.** This process focuses on visualizing and generating sign language gestures by translating gloss into sign language expressions represented as image frames facilitating the G2S transformation. The output can be an avatar that functions as a dynamic representation of the signs and facial expressions to close the communication gap between spoken and sign languages (Huang et al. 2022).

SLMT systems are mission-critical since they can be utilized during emergencies and disastrous events (Tonkin 2017). Therefore, the following requirements are essential during translation:

- **Large datasets for accurate translation:** SLMT systems require large and diverse datasets to train the AI models and achieve accurate translation effectively. As sign languages exhibit regional variations and individual signing styles (Jay 2011), a large dataset ensures that the model captures the richness of the signing patterns. In addition, the dataset should include a wide range of vocabulary, expressions, and contexts to enhance the system's ability to translate various sentences with the intended context accurately. An efficient and accurate system must recognize and understand the sign language grammar, facial expressions, and body movements (Jay 2011). Achieving high accuracy involves linguistic accuracy and capturing the context embedded in sign languages, which can be distinct from spoken languages.

- **Real-time translation:** Real-time translation is essential for SLMT systems to provide a seamless and uninterrupted user experience (Ismail et al. 2014). To achieve real-time translation, the SLMT system should satisfy the Quality-of-Service (QoS) requirements such as ultra-low latency and high data rate, for instance in a Metaverse world (Ismail and Buyya 2023). This requirement relies on model optimization, efficient computing resources, and lightweight algorithms to ensure rapid and immediate translation (Ismail et al. 2020; Al-Doghman et al. 2023).
- **Privacy and security measurements:** Language translation often involves personal information being translated. A robust SLMT system must consider the privacy and security standards during a human-MT interaction. This includes the security of user data, encryption during translation, and user consent for any data collection (Nunes Vieira et al. 2023; Bhuiyan et al. 2021). Compliance with privacy regulations and implementing ethical practices are essential to build trust among users.

6 Sign language datasets

We present a comprehensive list of the public SLMT in Table 2. Each dataset has essential attributes, including the year, sign language, number of signers (#Signers), number of videos (#Videos), resolution, and acquisition mode. These datasets have different environments, qualities, constraints, and complexities.

SLMT datasets cover different sign languages, including American (Zahedi et al. 2005; Dreuw et al. 2008; Athitsos et al. 2008; Johnston 2010; “American Sign Language Dataset”. 2020; Ananthanarayana et al. 2021; Duarte et al. 2021; Joze and Koller 2018; Hassan et al. 2020; Dongxu et al. 2020; Mavi and Dikle 2022; Othman et al. 2012; Mavi 2020; Thangali et al. 2011; Pugeault and Bowden 2011; Kurakin et al. 2012), Arabic (Al-Hammadi et al. 2020; Sidig et al. 2021; Al-Barham et al. 2301; Dahmani and Larabi 2014; Latif et al. 2019), Australian (Johnston 2010), Brazilian (Cerna et al. 2021; Almeida et al. 2019), British (Schembri et al. 2013; Matthes et al. 2012), Chinese (Zhou et al. 2021; Chai et al. 2014; Guo et al. 2018; Huang et al. 2018; Zhang et al. 2016; Pu et al. 2016), Columbian (Rodriguez et al. 2020), Finnish (Viitaniemi et al. 2014), French (Matthes et al. 2012), German (Camgoz et al. 2018; Matthes et al. 2012; Koller et al. 2015; Dreuw et al. 2006; Cooper et al. 2012), Greek (Matthes et al. 2012; Adaloglou et al. 2022), Indian (Elakkiya and Natarajan 2021; Sridhar et al. 2020), Irish (Oliveira et al. 2017), Korean (Ko et al. 2019; Yang et al. 2020), Persian (Rastgoo et al. Jul. 2020), Polish (Oszust and Wysocki 2013), Russian (Kapitanov et al. 2305), and Turkish (Sincan and Keles 2020; Camgoz et al. 2016). There is a preference among researchers for developing datasets using Continuous Sign Language Recognition (CSLR) over Isolated Sign Language Recognition (ISLR). This aligns with the practical relevance of CSLR in real-world scenarios, where sign language communication has a continuous and dynamic nature (Jay 2011). Moreover, the resolution and acquisition methods vary for some datasets. These variations introduce more diversity and complexity, potentially making them valuable resources for addressing challenging research tasks.

Figure 6 presents the frequency and average number of videos for the public continuous sign language datasets. The figure shows that even though American Sign Language (ASL) dominates other sign languages in frequency, it has one of the lowest average numbers of videos (1106). The dominance of ASL is due to its popularity (Ethnologue 2023). Hence, the availability of its resources and heightened research interest. On the other hand, the figure

Table 2 Public sign language datasets

Dataset	Year	Sign Language	Video	Gloss	Text	#Signers	#Videos	Resolution	Acquisition	Type
BOSTON-50 (Zahedi et al. 2005)	2005	American	✓	✓	✗	3	483	195×165	RGB	CSLR
BOSTON-104 (Dreu et al. 2008)	2008	American	✓	✓	✗	3	201	321×242	RGB	
ASLVD (Athitsos et al. 2008)	2008	American	✓	✓	✗	4	NR	Varies	RGB	
Auslan dataset (Johnston 2010)	2010	Australian	✓	✓	✗	100	1100	NA	NA	
ASLG-PG12 (Othman et al. 2012)	2012	American	✗	✓	✓	NA	NA	NA	NA	ISLR
BSL Corpus (Schembri et al. 2013)	2013	British	✓	✓	✓	249	NR	NR	RGB	
Devisign (Chai et al. 2014)	2014	Chinese	✓	✓	✗	8	24,000	NR	RGB, Depth	
S-pot (Viitaniemi et al. 2014)	2014	Finnish	✓	✓	✓	5	5539	720×576	RGB (Betacam)	
Phoenix-2014 (Koller et al. 2015)	2015	German	✓	✓	✓	9	6841	210×260	RGB	
BosphorusSign (Camgoz et al. 2016)	2016	Turkish	✓	✓	✗	10	NR	1920×1080	Kinect v2	
CSL (Guo et al. 2018)	2018	Chinese	✓	✓	✗	50	25,000	1920×1080	RGB, depth camera	
USTC-CSL (Huang et al. 2018)	2018	Chinese	✓	✓	✗	50	25,000	1920×1080	HD RGB	
PHOENIX-2014T (Camgoz et al. 2018)	2018	German	✓	✓	✓	9	8257	210×260	RGB	
KETI (Ko et al. 2019)	2019	Korean	✓	✓	✓	10	14,672	1920×1080	HD RGB	
ASLLRP ("American Sign Language Dataset" 2020)	2020	American	✓	✓	✓	7	NR	Varies	RGB	
CoL-SLTD (Rodriguez et al. 2020)	2020	Columbian	✓	✓	✗	13	1020	448×448	RGB	
ASLing (Ananthanarayana et al. 2021)	2021	American	✓	✗	✓	7	1284	450×600	RGB	
How2Sign (Duarte et al. 2021)	2021	American	✓	✓	✗	11	2456	1280×720	RGB	
LIBRAS-UFOP (Cerna et al. 2021)	2021	Brazilian	✓	✓	✗	5	3040	640×480	Kinect	
The GSL Dataset (Adaloglou et al. 2022)	2021	Greek	✓	✓	✓	7	10,295	840×480	Intel RealSense	
ISL-CSLTR (Elakkia and Natarajan 2021)	2021	Indian	✓	✓	✓	7	700	NR	RGB	
RTWH Fingerspelling (Dreu et al. 2006)	2006	German	✓	✓	✗	20	1400	Varies	Webcam and camcorder	
Boston ASL LVD (Thangali et al. 2011)	2011	American	✓	✓	✗	6	3300	90×90	NR	
ASL Fingerspelling A (Pugeault and Bowden 2011)	2011	American	✓	✓	✗	4	48,000	NR	RGB	
MSR (Kurakin et al. 2012)	2012	American	✓	✓	✗	10	12	640×480	Kinect	

Table 2 (continued)

Dataset	Year	Sign Language	Video	Gloss	Text	#Signers	#Videos	Resolution	Acquisition	Type
Dicta-Sign (Matthes et al. 2012)	2012	British, German, Greek, and French	✓	✓	✗	14–16/ SL	+ 4000	NA	RGB	
DGS Kinect 40 (Cooper et al. 2012)	2012	German	✓	✓	✗	15	40	640×480	Kinect	
PSL Kinect 30 (Oszust and Wysocki 2013)	2013	Polish	✓	✓	✗	1	30	640×480	Kinect	
PSL ToF 84 (Oszust and Wysocki 2013)	2013	Polish	✓	✓	✗	1	84	640×480	Kinect	
Alphabets-Uniform (Dahmani and Larabi 2014)	2014	Arabic	✓	✓	✗	24	2800	NR	NR	
Alphabets-Complex (Dahmani and Larabi 2014)	2014	Arabic	✓	✓	✗	8	960	NR	NR	
CSL-500 (Pu et al. 2016)	2016	Chinese	✓	✓	✗	50	125,000	1920×1080	RGB, depth camera	
SLR-100 (Zhang et al. 2016)	2016	Chinese	✓	✓	✗	50	5000	1280×720	Kinect	
ISL-HS (Oliveira et al. 2017)	2017	Irish	✓	✓	✗	6	468	640×480	RGB	
MS-ASL (Joze and Koller 2018)	2019	American	✓	✓	✗	222	25,513	Varies	RGB	
ArSL2018 (Latif et al. 2019)	2019	Arabic	✓	✓	✗	40	54,049	64×64	Mobile camera	
MINDS-Libras (Almeida et al. 2019)	2019	Brazilian	✓	✓	✗	12	1200	1080×1920	Kinect	
ASL-100-RGBD (Hassan et al. 2020)	2020	American	✓	✓	✗	22	100	1080×1920	Kinect	
WLASL (Dongxu et al. 2020)	2020	American	✓	✓	✗	119	21,013	Varies	RGB	
Sign Language Digits (Mavi 2020)	2020	American	✓	✓	✗	218	21,800	100×100	RGB	
KSU-SSL (Al-Hammadi et al. 2020)	2020	Arabic	✓	✓	✗	40	16,000	Varies	RBG and Kinect	
INCLUDE (Sridhar et al. 2020)	2020	Indian	✓	✓	✗	7	4287	1920×1080	HD RGB	
KSL (Yang et al. 2020)	2020	Korean	✓	✓	✗	20	1540	1280×720	RGB	
RKS-PERSIANSIGN (Rastgoo et al. 2020)	2020	Persian	✓	✓	✗	10	100	NR	RGB	
AUTSL (Sincan and Keles 2020)	2020	Turkish	✓	✓	✗	43	38,336	512×512	Kinect	

Table 2 (continued)

Dataset	Year	Sign Language	Video	Gloss	Text	#Signers	#Videos	Resolution	Acquisition	Type
KArSL (Sidig et al. 2021)	2021	Arabic	✓	✓	✗	3	75,300	Varies	Kinect	
27 Class ASL (Mavi and Dikle 2022)	2022	American	✓	✓	✗	173	22,490	3024 × 3024	HD camera	
AASL (Al-Barham et al. 2301)	2023	Arabic	✓	✓	✗	200	7856	Varies	RGB	
Solvo (Kapitanov et al. 2305)	2023	Russian	✓	✓	✗	194	20,400	1920 × 1080	HD RGB	

NA not applicable; NR not reported

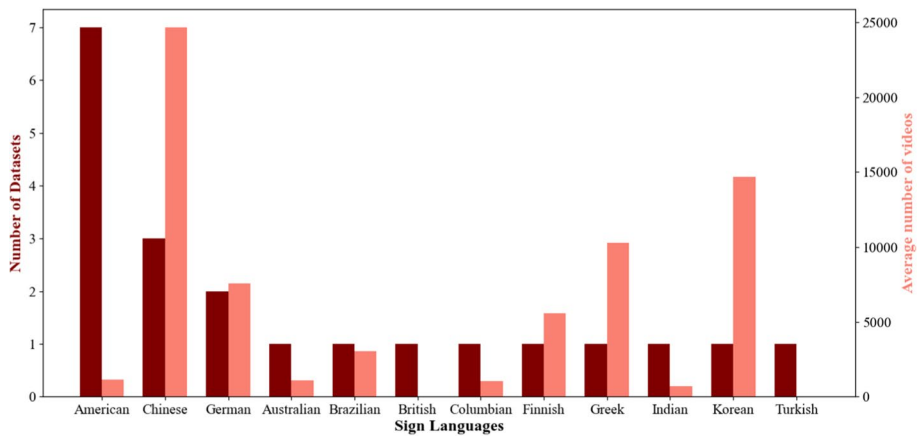


Fig. 6 Frequency and average videos for continuous sign languages based on the public datasets

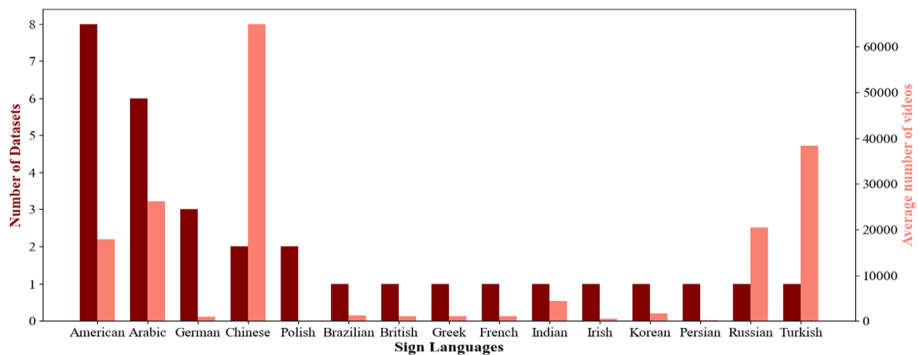


Fig. 7 Frequency and average videos for isolated sign languages based on the public datasets

shows that Chinese Sign Language has the largest average number of videos (24,667). Figure 7 presents the frequency and average number of videos for the public isolated sign language datasets. Like the previous figure, ASL dominates other sign languages in frequency, although it has an average number of videos of 17,779, which is the fifth largest sign language. The figure also shows that the Chinese Sign Language has the largest average number of videos (65,000). Moreover, Fig. 8 illustrates the coverage of components of the public datasets, showing that most public SLMT datasets cover video frames and gloss, followed by datasets that cover all three components and a few datasets that cover video frames and text. This indicates that the direct S2T translation is not as popular as S2G or S2G2T, highlighting the importance of gloss representation in this process.

In summary, German Sign Language has attracted more attention due to the implementation of Phoenix-2014 (Koller et al. 2015) dataset and its extension Phoenix-2014T (Camgoz et al. 2018). Based on the popularity of PHOENIX-2014T dataset, we suggest utilizing it when introducing new models to ensure accurate comparisons with existing literature until a more extensive and diverse dataset is available. We also suggest creating a large continuous SLMT multilingual dataset to aid in building one generative multilingual SLMT model.

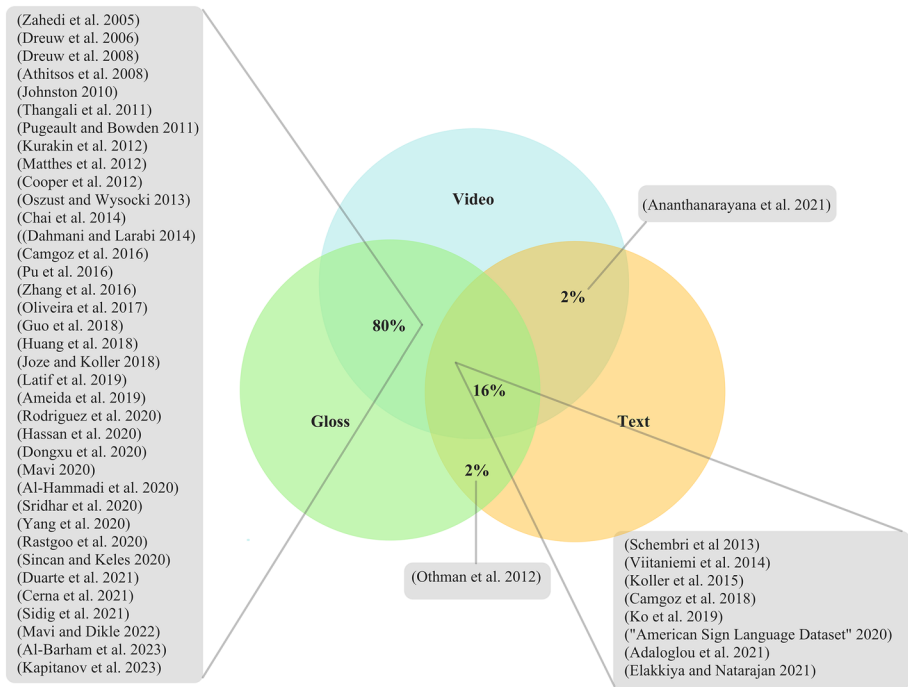


Fig. 8 Coverage of components of the public datasets

7 Performance evaluation metrics

The assessment of translation systems is essential for understanding their effectiveness and accuracy. This section describes the performance metrics used in SLMT which are Bilingual Evaluation Understudy (BLEU) (Papineni et al. 2001), Recall-Oriented Understudy for Gisting Evaluation (ROUGE) (Lin 2004), Word Error Rate (WER) (Klakow and Peters 2002), and Accuracy.

7.1 Bilingual evaluation understudy (BLEU)

BLEU is used to measure the similarity between the machine and human translations of sign language to spoken language and vice-versa. It focuses on the precision of n-grams (Papineni et al. 2001). BLEU scores are expressed on a scale from 0 to 1, where 1 indicates a perfect match with the reference translation.

This metric is divided into BLEU-1, BLEU-2, BLEU-3, and BLEU-4. These variants evaluate the concordance of the respective n-grams between the machine and human translations. This allows the evaluation of the lexical accuracy and appropriateness of the translations.

BLEU score is calculated using Eqs. (1) and (2).

$$BLEU = BP \cdot e^{\sum_{n=1}^N w_n \log(p_n)} \quad (1)$$

where p_n is the precision of n-grams, calculated as the ratio of the number of matching n-grams in the translation to the total number of n-grams in the translation, w_n is the weight of each n-gram size, and BP is the Brevity Penalty.

$$BP = \begin{cases} 1, & \text{if } c > r \\ e^{(1-\frac{r}{c})}, & \text{if } c \leq r \end{cases} \quad (2)$$

where c is the length of the candidate machine translation and r is the reference corpus length.

7.2 Recall-oriented understudy for gisting evaluation (ROUGE)

ROUGE assesses the overlap of n-grams, word sequences, and word pairs between the machine and human translations. Unlike BLEU, which primarily focuses on precision, ROUGE gives insight into recall, measuring how many of the human's n-grams are captured by the machine-translated text. This metric helps in understanding the extent to which important information is retained in the translation (Lin 2004). ROUGE is calculated using the following Eqs. (3–5).

$$Rouge = 2 \cdot \frac{Rouge_{recall} \cdot Rouge_{precision}}{Rouge_{recall} + Rouge_{precision}} \quad (3)$$

$$Rouge_{recall} = \frac{C}{M} \quad (4)$$

$$Rouge_{precision} = \frac{C}{G} \quad (5)$$

where C is the count of overlapping n-grams between the machine translation and the human translations. M is the total count of n-grams in the human translations. G is the total count of n-grams in the machine translations.

7.3 Word error rate (WER)

WER measures the translation by comparing the machine-translated against the human-translated text on a word-by-word basis (Klakow and Peters 2002). It calculates the number of substitutions (S), deletions (D), and insertions (I) needed to change the machine-translated text into the human-translated text, normalized over the number of words in the human translation (N), as shown in Eq. (6).

$$WER = \frac{S + D + I}{N} \quad (6)$$

7.4 Accuracy

Accuracy is particularly used for tasks that involve classification, such as identifying specific signs in SLTM systems. It is calculated by Eq. (7). However, (Li et al. 2022) used Eq. (8) to measure the translation accuracy.

$$\text{Accuracy} = \frac{\text{Number of correct translations}}{\text{Total number of translations}} \quad (7)$$

$$\text{Accuracy} = (1 - \text{WER}) \times 100 \quad (8)$$

In summary, the evaluation of SLMT systems through a variety of metrics—BLEU, ROUGE, WER, and Accuracy—offers a perspective on system performance, including aspects of precision, recall, error rate, and overall correctness. Together, these metrics enable comprehensive and nuanced assessments of SLMT systems, guiding researchers and developers in identifying strengths, weaknesses, and opportunities for advancements in the field.

8 Taxonomy of sign language machine translation algorithms

Figure 9 shows a proposed taxonomy based on a retrospective analysis of the temporal evolution of the SLMT algorithms. We classify these algorithms into four categories: (1) RBMT (Kouremenos et al. 2018; Luqman and Mahmoud 2019; Costa et al. 2019), (2) EBMT (Morrissey and Way 2005), (3) SMT (Ibrahim et al. 2018; Klomsae et al. 2017; Boulares and Jemni 2016; Bhagwat et al. 2021), and (4) NMT (Camgoz et al. 2020, 2018; Li and Meng 2022; Huang et al. 2022; Li et al. 2022; Zhou et al. 2021, 2022; Chen et al. 2022; Guo et al. 2018; Rodriguez et al. 2020; Alves et al. 2022; Farooq et al. 2023; Stoll et al. 2020; Zhang et al. 2023; Kan et al. 2022; Yin et al. 2021, 2022; Fang et al. 2017; Zheng et al. 2020; Castro et al. 2023; Zhao et al. 2022; Zheng et al. 2021; Guo et al. 2020; Khan et al. 2021; Jin et al. 2022; Kim and Baek 2023; Chaudhary et al. 2022; Tang et al. 2022; Natarajan and Elakkiya 2022). We map each of these algorithms to the corresponding framework sign language translation stages that were investigated in the literature. We also present a comparison between the SLMT algorithms in the literature in terms of the sign language considered, the dataset(s) used, the translation algorithms and the feature extraction techniques employed, and the corresponding performance, as shown in Table 3.

8.1 Rule-based machine translation (RBMT)

Sign language RBMT relies on grammatical rules, dictionaries, and syntactic and semantic analysis for translation (Kay 1984). Few works implemented RBMT for sign language translation. These works are divided into two categories: (1) T2G (Kouremenos et al. 2018; Luqman and Mahmoud 2019), and (2) S2T (Costa et al. 2019). Regarding T2G translation (Kouremenos et al. 2018; Luqman and Mahmoud 2019), authors followed a three-step rule-based approach which constitutes of Part-of-Speech (POS) tagging, where every word is mapped to its corresponding type, such as noun, verb, and adjectives (Charniak et al. 1996), chunk partial parser, where each sentence is chunked into sub-sentences (Abney 1992), chunk transfer on the sub-sentence level (Kouremenos et al. 2018) along with Morpho transfer on the word level (Sheridan and Smeaton 1992) to produce the gloss. (Kouremenos et al. 2018) applied the approach on a Greek dataset while (Luqman and Mahmoud 2019) applied the same approach on an Arabic dataset of a smaller size, which resulted in a precision of 84 and 35 respectively.

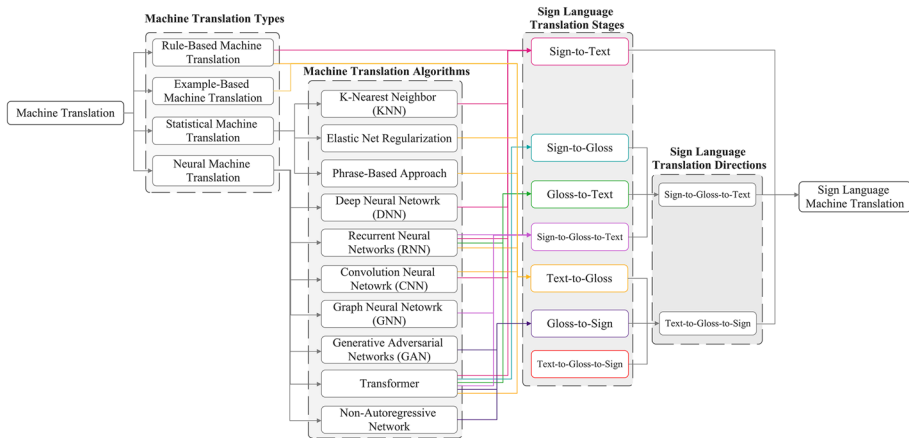


Fig. 9 Sign Language Translation Algorithms Taxonomy

On the other hand, in S2T translation, (Costa et al. 2019) followed the three-step rule-based approach on Brazilian sign language videos and obtained a precision of 21.1. *Rule1* shows an example of a grammatical rule in sign language (Luqman and Mahmoud 2019), where S is the Subject, O is the Object, and V is the Verb.

IF the input is an Arabic sentence with structure VOS, THEN reorder words to SVO (Rule1).

In summary, RBMT precision is language-dependent as it is high in Greek and low in Arabic. In addition, the algorithm may face challenges when dealing with dialectal variations of sign languages, expressions, and context-dependency (Kay 1984).

8.2 Example-based machine translation (EBMT)

EBMT was introduced for language translation by looking at the similarity of the corresponding corpus. Despite its precise translations, EBMT has limited translation coverage because no corpus captures all the linguistic nuances (Wang et al. 2022). In 2005, (Morrissey and Way 2005) presented an example-based approach to translating English text to Dutch gloss by searching for the best matches, based on word occurrences, POS labels, and bilingual dictionaries, and recombining relevant parts of the translated sentence using closed-class words (Gough and Way 2004) as markers to segment and align source and target sentences. Despite its precision, this method does not scale with a corpus increasing size (Katwe et al. 2023).

8.3 Statistical machine translation (SMT)

MT models shifted from EBMT to SMT due to the limitations faced by the former type and the need for a more efficient and scalable translation. SMT can solve these issues as it relies on large corpora and probabilistic methods (Lopez 2008) and handles ambiguity better than EBMT. The following are the SMT approaches used in SLMT:

Table 3 Comparison between Sign Language Machine Translation Algorithms

Translation	Work	Sign Language	Dataset	Translation Model		Performance Metrics							
				Algorithm	Feature Extraction	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	WER	Accuracy	Execution Time
Rule-Based Machine Translation (RBMT)													
Text-to-Gloss	Kouremenos et al. (2018)	Greek	1015 sentences	Rule-based	POS tagging and chunk partial parser	90	NR	NR	84	NR	NR	NR	NR
		Arabic	600 sentences		Morphological and syntactic analysis	NR	NR	NR	35	NR	0.6	NR	NR
Sign-to-Text	Costa et al. (2019)	Brazilian	69 sentences		POS tagging and text parsing	64.3	42.1	29.4	21.1	NR	58.3	NR	NR
Example-Based Machine Translation (EBMT)													
Text-to-Gloss	Morrissey and Way (2005)	Dutch	561 sentences	EBMT	POS tagging	NR	NR	NR	NR	NR	NR	NR	NR
Statistical Machine Translation (SMT)													
Sign-to-Text	Ibrahim et al. Oct. (2018)	Arabic	30 words	Euclidean Distance	Dynamic skin detector, skin-blob tracking, and cascade boosting algorithm	NR	NR	NR	NR	NR	NR	97	NR
Text-to-Gloss	Klomsae et al. (2017)	Thai and American	10 words for TSL and 31 words for ASL from BOSTON-50*	Fuzzy KNN	Not reported	NR	NR	NR	NR	NR	NR	91.4	NR
		American	300 sentences	Elastic Net Regularization	n-spectrum weighted word kernel	28	NR	NR	NR	NR	NR	NR	5.9 s/word
Text-to-Gloss	Boulares and Jenni (2016)	Marathi	1200 sentences	Phrase-based	Tokenization, POS tagging, and phrase identification	NR	NR	NR	NR	NR	NR	NR	NR

Table 3 (continued)

Translation	Work	Sign Language	Dataset	Translation Model		Performance Metrics							
				Algorithm	Feature Extraction	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	WER	Accuracy	Execution Time
Neural Machine Translation (NMT)													
Sign-to-Gloss	Cangoz et al. (2020)	German	PHOENIX-2014T	Transformer	3D ResNets	NR	NR	NR	NR	NR	24.6	NR	NR
	Zhang et al. (2023)				Self-Mutual Knowledge Distillation model	NR	NR	NR	NR	NR	19.2	NR	NR
Gloss-to-Text	Cangoz et al. (2020)			Transformer		48.9	36.9	29.5	24.5	NR	NR	NR	NR
	Cangoz et al. (2018)			GRU with attention layer** and LSTM with attention layer		44.1	31.5	23.9	19.3	45.5	NR	NR	NR
	Huang et al. (2022)			Transformer	Fast Fourier Transform	33.7	24.8	19.2	15.8	36.5	NR	NR	NR
Text-to-Gloss	Chen et al. (2022)				mBART's SentencePiece	52.7	40	32.1	26.7	52.5	NR	NR	NR
	Huang et al. (2022)	German	PHOENIX-2014T	Transformer	Fast Fourier Transform	47.9	NR	NR	14.8	51.2	NR	NR	NR
	Alves et al. (2022)	Portuguese	150k sentences	LSTM** and BiLSTM	Morphological Analysis	95.6	93.4	91.5	85.1	NR	NR	NR	NR
	Farooq et al. (2023)	Pakistani	50k sentences	GRU** BiGRU, BiLSTM, and LSTM with attention layer	NR	83.3	NR	NR	51.4	NR	0.2	NR	NR
Gloss-to-Sign	Stoll et al. (2020)	German	PHOE-NIXNR2014T	GRU	NR	50.7	32.3	21.5	15.3	48.1	4.5	NR	NR
	Huang et al. (2022)			Non-autoregressive network	NR	15.6	NR	NR	7.2	19.1	83.3	NR	NR
	Stoll et al. (2020)			GAN	NR	NR	NR	NR	NR	NR	NR	NR	NR
	Tang et al. (2022)			Transformer		23.1	14.2	10.5	8	23.5	81.8	NR	NR
	Natarajan and Elakkiya (2022)	German, Indian and American	PHOENIX-2014T, ISL-CSLTR, and UCF101	GAN	NR	NR	NR	NR	NR	NR	NR	NR	NR

Table 3 (continued)

Translation	Work	Sign Language	Dataset	Translation Model		Performance Metrics							
				Algorithm	Feature Extraction	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	WER	Accuracy	Execution Time
Sign-to-Gloss-to-Text	Cangoz et al. (2020)	German	PHOENIX-2014T	Transformer		48.5	35.4	27.6	22.5	NR	NR	NR	NR
	Cangoz et al. (2018)			GRU with attention layer	NR	43.3	30.4	22.8	18.1	43.8	NR	NR	NR
	Li and Meng (2022)	Chinese and German	SLR-100, PHOENIX-2014T, and CSL daily*	Transformer	Multi-View Spatial–Temporal Embedding Network	45.4	32.5	25.1	20.4	NR	NR	NR	NR
	Zhou et al. (2021)	Chinese and German	CSL-Daily & PHOENIX-2014T*		NR	48.6	36.1	28.5	23.5	49.4	NR	NR	NR
	Chen et al. (2022)	American, German, and Chinese	Kinetics-400 and WLASL for pretraining, PHOENIX-2014T and CSL-Daily* for experiments	mBART's SentencePiece		50.3	37.4	28.1	21.5	51.4	NR	NR	NR
	Zhang et al. (2023)	Chinese and German	CSL-Daily* & PHOENIX-2014T	Transformer	SMKD model	54.4	40.3	30.5	23.8	53.1	NR	NR	NR
	Kan et al. (2022)		CSL-Daily, and PHOENIX-2014T*	Hierarchical GNN	Hierarchical spatiotemporal graphs	45.2	34.7	27.1	22.3	NR	19.8	NR	NR
	Yin et al. (2021)	German	PHOENIX-2014T	Transformer	CNN	48.2	35.6	28	23.1	49.2	NR	NR	NR
	Zhou et al. (2022)	Chinese and German	CSL, PHOE-NIX-2014, & PHOENIX-2014T*	BiLSTM	SMC and TM modules	47	36.1	28.7	23.7	46.7	NR	NR	NR
Sign-to-Text	Cangoz et al. (2020)	German	PHOENIX-2014T	Transformer	46.6	33.7	26.2	21.3	NR	26.2	NR	NR	NR

Table 3 (continued)

Translation	Work	Sign Language	Dataset	Translation Model		Performance Metrics							
				Algorithm	Feature Extraction	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	WER	Accuracy	Execution Time
	Li and Meng (2022)	German and Chinese	SLR-100, PHOENIX-2014T*, and CSL-Daily	Transformer	Multi-View Spatial–Temporal Embedding Network, 3D CNN	49.6	36.5	29.1	22.5	NR	23.2	NR	NR
	Li et al. (2022)	American	69 words used in 41 sentences	BiLSTM	NR	NR	NR	NR	NR	NR	NR	95	1.4 s/sign
	Zhou et al. (2021)	German and Chinese	CSL-Daily & PHOENIX-2014T*	Transformer	NR	50.8	37.8	29.7	24.3	49.5	23.9	NR	NR
	Chen et al. (2022)	German, and Chinese	PHOENIX-2014T*, CSL-Daily	Transformer	mBART's SentencePiece	54	41.8	33.8	28.4	52.7	NR	NR	NR
	Guo et al. (2018)	Chinese	CSL	hierarchical-LSTM with attention layer	3D CNN	50.8	33	20.7	NR	50.3	0.6	NR	NR
	Rodriguez et al. (2020)	Columbian	CoL-SLTD	LSTM	3D CNN	30.1	12.9	7.1	4.7	30.6	88.9	NR	NR
	Fang et al. (2017)	American	7,306 samples covering 56 words and 100 sentences	Hierarchical BiDNN with LSTM	Leap motion sensors	NR	NR	NR	NR	NR	NR	94.5	NR
	Zheng et al. (2020)	German	PHOENIX-2014T	BiGRU	Frame stream density compression and CNN	32.2	19.4	13.7	10.7	32.3	NR	NR	NR
	Castro et al. (2023)	Brazilian, Indian, and Korean	MINDS-Libras*, INCLUDE, KSL, and LIBRAS-UFOF	3D CNN	NR	NR	NR	NR	NR	NR	NR	91	NR
	Zhao et al. (2022)	Chinese and German	CSL Daily and PHOENIX-2014T*	Transformer	CNN	36.7	25.4	18.9	15.2	38.9	0.7	NR	NR
	Zheng et al. (2021)	German	PHOENIX-2014T	GRU	Semantic Focus of Interest Network with Face Highlight Module	NR	NR	NR	10.9	34.9	NR	NR	NR

Table 3 (continued)

Translation	Work	Sign Language	Dataset	Translation Model		Performance Metrics							
				Algorithm	Feature Extraction	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	WER	Accuracy	Execution Time
	Guo et al. (2020)	Chinese	CSL-Daily	LSTM with attention layer	3D CNN	45	23.8	12.7	NR	44.9	67.2	NR	NR
	Khan et al. (2021)	Malaysian	19 signs	CNN with attention layer	CNN	NR	NR	NR	NR	NR	NR	99.4	NR
	Jin et al. (2022)	Chinese and German	CSL-Daily and PHOENIX-2014T*	Transformer	CNN	47.2	NR	NR	19.7	46.2	NR	NR	NR
	Yin et al. (2022)	Multi-language	SP-10	Transformer	NR	NR	NR	NR	4.7	32.7	NR	NR	NR
	Kim and Baek (2023)	German and Korean	PHOENIX-2014T and KETI*	GRU	AlphaPose, Normalization, Stochastic Augmentation, and Skip Sampling	NR	NR	NR	85.5	84.9	NR	NR	NR
	Chaudhary et al. (2022)	American and German	ASLing and PHOENIX-2014T*	Transformer	CNN, OpenPose, and Optical Flow embeddings	39.2	24.6	16.9	12.3	NR	NR	NR	NR

*Best dataset performance. **Best algorithm performance

NR not reported

- *Elastic Net Regression*: In 2016, (Boulares and Jemni 2016) translated 300 ASL sentences using a combination of L1 (Lasso) and L2 (Ridge) penalties of the lasso and ridge method. First, they applied feature mapping to convert the sentences into vectors using the n-spectrum weighted word kernel technique. Then, they mapped between the source and target sentences, which resulted in a linear regression function. This algorithm solves the regression sensitivity and instability and avoids over-fitting (Zou and Hastie 2005). Here, L1 penalty penalizes a model based on the sum of the absolute coefficient values. It minimizes the size of all coefficients removing a particular feature of predicted translation from the model. Equation (9) shows the L1 penalty, where p is the number of input features, j is the input feature, and $|\beta_j|$ is the absolute value of the coefficient associated with the j th feature.

$$l1 = \sum_{j=0}^p |\beta_j| \quad (9)$$

On the other hand, the L2 penalty penalizes models based on the sum of squared coefficient values, as shown in Eq. (10). Where β_s^2 is the square of the coefficient associated with the j th feature.

$$l2 = \sum_{j=0}^p \beta_j^2 \quad (10)$$

Elastic Net Regularization introduces two hyperparameters: alpha (α) and lambda (λ). Alpha controls the balance between L1 and L2 penalties, with values between 0 and 1, while lambda controls the overall strength of the combined penalty. Equations (11) and (12) present the elastic net penalty and loss formulas, respectively. The loss here is usually a mean square error between the predicted and actual translations.

$$\text{Elastic net penalty} = (\alpha \cdot l1) + ((1 - \alpha) \cdot l2) \quad (11)$$

$$\text{Elastic net loss} = \text{loss} + (\lambda \cdot \text{Elastic net penalty}) \quad (12)$$

- *K-Nearest Neighbors (KNN)*: In 2017, (Klomsae et al. 2017) applied Fuzzy-KNN (FKNN), an extension of KNN, to find the matched sign words in Thai Sign Language and ASL S2T translation. To calculate Fuzzy KNN, the authors used Euclidean Distance (Tabak 2014) in the key point matching process to compare the distance between the two nearest neighbor key points in the dataset and the current key point. Then they calculated the membership value of each data point belonging to multiple words in spoken language using the Gaussian Function (Zadeh 1965), as shown in Eq. (13). This function describes the degree of belonging of a data point to a particular word in the spoken language. Based on this, the authors assign each sign gesture to a spoken language word, as shown in Eq. (14).

$$w_{ij} = \exp\left(-\frac{d_{ij}^2}{2z^2}\right) \quad (13)$$

where w_{ij} is a fuzzification function, and z is a parameter controlling the width of the membership function.

$$FuzzyClass = \frac{\sum_{i=1}^k w_{ij} \cdot y_i}{\sum_{i=1}^k w_{ij}} \quad (14)$$

where y_i the class label of the i -th sample.

- *Phrase-Based Approach*: In 2021, (Bhagwat et al. 2021) applied the phrase-based approach to translate Marathi text to Indian gloss by applying tokenization and POS tagging, then identifying the phrases' structure using a knowledge analyzer that extracts the syntactical information such as sentence structure and verb tense, which is translated into gloss.

In summary, SMT overcomes the challenges faced by EBMT by relying on large corpora and probabilistic methods, aligning words and phrases across both languages and applying statistical formulas to generate translations (Lopez 2008), thus handling ambiguity better than EBMT. In addition, SMT is more scalable and cost-effective, as it requires fewer manual efforts to maintain the linguistic rules. The SMT methodologies applied in SLMT (Ibrahim et al. 2018; Klomsae et al. 2017; Boulares and Jemni 2016) showcase the different approaches in MT and highlight advancements in linguistic and mathematical techniques.

8.4 Neural machine translation (NMT)

NMT utilizes deep learning algorithms to translate between languages. Its components include neural network structures, datasets containing original and translated sentences, semantic representations through embeddings, encoder-decoder frameworks, attention mechanisms, and training processes involving backpropagation and gradient descent (Sutskever et al. 2014; Bahdanau et al. 1409). These features give superiority to NMT over SMT as they allow the algorithm to handle complex language structures and capture long-range dependencies and language patterns. However, NMT still faces challenges, such as the need for larger datasets and the unique linguistic and spatial characteristics of sign languages.

- *Deep Neural Network (DNN)*: In 2017, (Fang et al. 2017) captured more than 7000 isolated and continuous ASL signs from a private dataset and extracted their features using a leap motion sensor. The authors then translated the signs using a Hierarchical BiDNN with LSTM in S2T translation and obtained a 94.5% translation accuracy. The DNN algorithms they applied consist of neurons, weights, biases, and functions (Devan and Khare 2020). Initially, the input is fed into the input layer to be represented as vectors. Each neuron performs a weighted sum of inputs in the hidden layers followed by an activation function, as shown in Eq. (15). Lastly, the output layer generates the result, depending on the problem.

$$x_i = f\left(\sum_{j=1}^{N_i} (w_{ij} \cdot a_{i-1j} + b_i)\right) \quad (15)$$

where x_i is the weighted sum for neuron i in layer i . w_{ij} is the weight connecting neuron j in layer $i-1$ to neuron i in layer i . a_{i-1j} is the output (activation) of neuron j in layer $i-1$. b_i is the bias for neuron i in layer i . f is the activation function.

- **Recurrent Neural Networks (RNN):** The implementation of RNN in SLMT focuses on two primary architectures: Gated Recurrent Unit (GRU) and Long-Short-Term Memory (LSTM). Both architectures are well-known for their ability to manage memory through gating mechanisms, allowing for selective retention or forgetting of information. While they share this key principle, GRU and LSTM have major differences in their architectures. GRU involves a hidden state (h_t), reset (r_t) and update (u_t) gating mechanisms (Cho et al. 2014). h_t indicates the algorithm's memory from the previous time step and is updated at every time step, as shown in Eq. (16). r_t controls how much of the previous hidden state (h_{t-1}) should be forgotten by using the sigmoid (σ) activation function to produce values between 0 and 1 for each element in the hidden state, as shown in Eq. (17). u_t aims to control how much of the previous hidden state (h_{t-1}) should be passed and remembered in the future, as shown in Eq. (18). r_t and u_t affect the final translation through the current memory content (h'_t), which stores relevant past knowledge, as presented in Eq. (19). The final translation output (o_t) is calculated by Eq. (20).

$$h_t = \tanh(w_h * h_{t-1} + w_{hx} * x_t + b_h) \quad (16)$$

where $\tanh$ is the hyperbolic tangent activation function, w_h is the weight of the current hidden state, w_{hx} is the weight of the input sentence's hidden state, x_t is the vector of the original sentence, and b_h is the bias of the current hidden state.

$$r_t = \sigma(wr_{h-1} * h_{t-1} + wr_x * x_t) \quad (17)$$

where σ is the sigmoid activation function, wr_{h-1} and wr_x are the weights of the previous hidden state and the input sentence respectively, in the reset gate.

$$u_t = \sigma(wu_{h-1} * h_{t-1} + wu_x * x_t) \quad (18)$$

where wu_{h-1} and wu_x are the weights of the previous hidden state and the input sentence respectively, in the update gate.

$$h'_t = \tanh(w_x * x_t + r_t * w_{h-1}) \quad (19)$$

where w_x is the weight of the input sentence, and w_{h-1} is the weight of the previous hidden state.

$$o_t = u_t * h_{t-1} + (1 - u_t) * h'_t \quad (20)$$

On the other hand, LSTM includes three gating mechanisms: forget (f_t), input (i_t), and output (o_t) gates. f_t removes that information that is no longer useful by taking the vector of the original sentence (x_t) and the previous hidden state (h_{t-1}) as inputs, as shown in Eq. (21). i_t adds valuable information to the cell while o_t controls which information should be used to produce the translation. Equations (22) and (23) represent the input and output gates respectively.

$$f_t = \sigma(wf_{h-1} * h_{t-1} + wf_x * x_t + b_f) \quad (21)$$

$$i_t = \sigma(wi_{h-1} * h_{t-1} + wi_x * x_t + b_i) \quad (22)$$

$$o_t = \sigma(wo_{h-1} * h_{t-1} + wo_x * x_t + b_o) \quad (23)$$

where w is the weight and b is the bias.

The application of GRU and LSTM in SLMT has led to significant developments and findings. For instance, in 2018, (Camgoz et al. 2018) applied GRU with a Luong attention mechanism for G2T and S2G2T translation showcasing enhanced performance on PHOENIX-2014T dataset compared to LSTM with an equivalent attention mechanism. GRU was also applied in T2G (Farooq et al. 2023; Stoll et al. 2020) and S2T (Zheng et al. 2020; Zheng et al. 2021; Kim and Baek 2023) translation. In T2G translation, (Farooq et al. 2023) applied the Bahdanau attention and tested it on translating English sentences to Pakistani Sign Language gloss while (Stoll et al. 2020) applied the Luong attention and tested it on PHOENIX-2014T dataset. Nevertheless, in S2T translation, (Zheng et al. 2020; Zheng et al. 2021) tested their proposed models on PHOENIX-2014T dataset while (Kim and Baek 2023) tested the architecture with the KETI and PHOENIX-2014T datasets. Despite KETI's smaller size, it yielded superior performance, suggesting differences in dataset characteristics or model adaptability need further investigation.

A comparison within the specific contexts of G2T and T2G translations on PHOENIX-2014T dataset illuminated the efficacy of these models, with (Stoll et al. 2020) showing superior results over (Camgoz et al. 2018). We believe that variance in performance stems from the distinct architectural parameters of the models.

Conversely, LSTM's application in SLMT highlights its ability to reduce the vanishing gradient problem and adeptly capture long-term dependencies within sequences, unlike GRUs (Hochreiter and Schmidhuber 1997). This is achieved through its distinct memory cell, setting it apart from GRU's approach. Few works applied LSTM in SLMT. In 2018, (Guo et al. 2018) proposed a framework consisting of 3D CNN for feature extraction and a hierarchical LSTM encoder-decoder model for S2T translation. The authors tested the model on their proposed CSL dataset. Other works also applied LSTM with varied feature extraction methods in the same translation manner (Li et al. 2022; Guo et al. 2018; Rodriguez et al. 2020; Guo et al. 2020). LSTM was also applied in T2G (Alves et al. 2022) and S2G2T translations (Zhou et al. 2022).

In conclusion, both GRU and LSTM architectures play essential roles in SLMT. The choice between GRU and LSTM often relies on the specific requirements of the translation task, including the need for memory efficiency, complexity of the sequence dependencies, and computational resources.

- *Convolution Neural Networks (CNN)*: CNN is a deep learning architecture used in image processing (Balas et al. 2020). It was implemented in sign video frame processing as in S2T translation in 2021 (Khan et al. 2021), where the authors proposed a model that consists of CNN with an attention layer and 2D ResNet, and tested the model on a private Malaysian Sign Language dataset. (Castro et al. 2023) applied 3D CNN for feature extraction and translation, utilizing three datasets for three sign languages: Brazilian, Indian, and Korean proposed a model that consists of CNN with an attention layer and 2D ResNet and tested the model on a private Malaysian Sign Language dataset. The algorithm consists of convolutional, pooling, and fully connected layers. The convolutional layer involves a kernel that slides over the input image and computes a dot product at each location, as shown in Eq. (24). On the other hand, the pooling layer reduces the spatial dimensions of the image, as shown in Eq. (25). The output from convolution and pooling layers is then passed through fully connected layers to produce the result. These fully connected layers are similar to the ones in DNN.

$$(I * K)(x, y) = \sum_i \sum_j I(x + i, y + j) \cdot K(i, j) \quad (24)$$

$$P(x, y) = \max (I(x', y')) \text{ for } x' \in [x, x + 2], y' \in [y, y + 2] \quad (25)$$

where $(I * K)(x, y)$ is the result of the convolutional operation at the x, y location. I is the input image, and K is the kernel. $P(x, y)$ is the output of the pooling layer at the location x, y .

Graph Neural Networks (GNN): In 2022, (Kan et al. 2022) implemented an encoder-decoder architecture to convert sign language videos into hierarchical spatiotemporal graph representations. The encoder uses graph convolution and a graph self-attention method to represent the sign videos in latent space, followed by hierarchical graph pooling. The decoder generates text using encoded reorientations in a hierarchical GNN. The algorithm captures the relationships between words or phrases. Here, the source and target sentences are represented in two different graphs. The words or phrases are the nodes, and the relationships are represented in edges (Scarselli et al. 2009).

- **Generative Adversarial Networks (GAN):** In 2020, (Stoll et al. 2020) applied GAN to produce sign videos from gloss input and evaluated the accuracy of the generated videos using qualitative measures. The generated videos' resolution and expressiveness were affected by the low resolution of the training data. (Natarajan and Elakkiya 2022) proposed a Dynamic GAN model to produce high-quality realistic sign language videos. The model consists of three main stages: (1) A GAN network to generate human-based sign gesture videos based on the provided gloss and ground truth images. (2) A CNN network to improve the quality of the generated samples by applying image alignment techniques. (3) A discriminator network to evaluate the quality of the generated. The model produced visually coherent and contextually relevant sign language gestures with high accuracy.
- **Transformers:** In 2020, (Camgoz et al. 2020) applied the transformer encoder-decoder architecture in S2G2T translation. Other works later applied the same architecture for the same translation (Li and Meng 2022; Zhou et al. 2021; Chen et al. 2022; Zhang et al. 2023; Yin et al. 2021). (Camgoz et al. 2020; Zhou et al. 2021; Yin et al. 2021), and tested the algorithm on PHOENIX-2014T dataset while (Li and Meng 2022; Chen et al. 2022; Zhang et al. 2023) tested it on CSL-Daily dataset. (Zhou et al. 2021) and (Zhang et al. 2023) outperformed the other works in each category, while (Zhang et al. 2023) surpassed (Zhou et al. 2021). We believe that this is due to the architecture parameters and the dataset size. In terms of S2G translation, (Camgoz et al. 2020) implemented an encoder-only transformer trained using a Connectionist Temporal Classification (CTC) loss to predict the gloss sequences. (Zhang et al. 2023) adopted a Self-Mutual Knowledge Distillation model based on CNN to extract the visual features of the sign videos and convert them to 1D temporal convolution, which in turn they use as an input to the encoder-decoder transformer to translate the sign videos into glosses. Both works tested the transformer model using PHOENIX-2014T dataset. When compared, (Zhang et al. 2023) outperforms (Camgoz et al. 2020) as it has a lower word error rate. On the other hand, in the G2T translation, (Huang et al. 2022) implemented an encoder-decoder transformer model variation that we call "encoder-decoder transformer fusion." This variation includes an FFT block and a gated bilateral fusion. However, the researchers omitted the residual connections and normalization layers. In addition, the decoders incorporated a length regulator to align the gloss and text sequences. (Chen et al. 2022) implemented a transformer encoder-decoder with bidirectional attention using mBART initialization. (Camgoz et al. 2020) employed a decoder-only transformer to translate from gloss to text as part of a S2G2T end-to-end framework. In this work, a direct comparison with an encoder-decoder transformer was not conducted. All these

works tested their architectures on PHEONIX-14 T dataset. Hence, when compared, we find that (Chen et al. 2022) outperformed the other two architectures, indicating that the encoder-decoder transformer architecture is the most precise. Other works applied transformer encoder-decoder in its original architecture in T2G (Huang et al. 2022), G2S (Huang et al. 2022; Tang et al. 2022), and S2T translations (Camgoz et al. 2020; Li and Meng 2022; Zhou et al. 2021; Chen et al. 2022; Zhang et al. 2023; Zhao et al. 2022; Jin et al. 2022; Yin et al. 2022; Chaudhary et al. 2022). (Chen et al. 2022) reported that their S2T approach significantly improved compared to S2G2T translation due to the visual pre-training using mBART. However, (Camgoz et al. 2020) found that S2T encountered challenges with long-term dependencies. Similarly, (Li and Meng 2022) and (Zhou et al. 2021) found that despite encountering a bottleneck in the form of gloss representation, the S2G2T approach outperformed S2T for the transformer architecture. (Chen et al. 2022) outperformed all architectures that tested the algorithm on PHOENIX-2014T dataset (Camgoz et al. 2020; Li and Meng 2022; Zhou et al. 2021; Chen et al. 2022; Zhang et al. 2023; Zhao et al. 2022; Jin et al. 2022; Chaudhary et al. 2022). We believe that this is due to the architecture's parameters. Regardless of the different types of transformers, the architecture's core component is the self-attention mechanism (Vaswani et al. 2017), which focuses on different parts of original sentences while generating the translated sentences.

Initially, the encoder-decoder transformer architecture was proposed by Vaswani et al. (2017) in 2017 for MT. The process of translation using this algorithm starts with tokenization to convert the original sentence into words or sub-words and build the model's vocabulary. This is followed by transforming these sentences into vectors through an embedding process. Then, these embeddings are encoded through positional encoding to determine the position of each word or the distance between different words in the sequence, as shown in Eqs. (26) and (27). The output of the positional encoding is then passed through the encoder, along with the embeddings, to produce a sequence of encoded representations that capture the relevant information in the sentence. The goal of adding positional encoding values to the embeddings is to provide meaningful distances between the embedding vectors once they are fed into the encoder. The encoder consists of multi-head self-attention, a feedforward network, and normalization and residual connections. The multi-head self-attention allows the model to weigh the importance of different parts of the input and capture relationships between words by computing weights for multiple sets of Query/Key/Value (Q/K/V) vectors, as presented in Eqs. (28) and (29). These vectors are obtained by multiplying the input embeddings by different weight matrices. On the other hand, the feedforward network adds non-linearities, helping to capture more complex patterns. Moreover, the normalization and residual connections stabilize and speed up the training of deep networks. The decoder of the transformer generates a sequential output using an autoregressive approach through the encoded representations and the previously generated words as input. These sequences are fed into a linear layer and then a SoftMax layer to output a set of probabilities for the generated word.

$$PE(pos, 2i) = \sin\left(\frac{pos}{1000^{\frac{2i}{d_{model}}}}\right) \quad (26)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{1000 \frac{2i}{d_{model}}}\right) \quad (27)$$

$$Attention = \text{Softmax}\left(\frac{QK}{\sqrt{d}}\right)V \quad (28)$$

$$MultiHead = \text{Concat}(Attention_1 \dots Attention_i) * w^o \quad (29)$$

where pos is the current position, i is the dimension index. d is the dimension. w^o is a learned weight matrix that adds the parameters in a single-head self-attention.

In summary, Transformers (Vaswani et al. 2017) altered the MT task, due to their ability to capture dependencies across sequences and solve the issues faced by previous models. However, this algorithm generates each token sequentially, as it is conditioned on the previously generated token sequence. This process is not parallelizable and is slow during inference. Nevertheless, different types of Transformers exist in SLMT systems, including encoder-only transformers, decoder-only transformers (like GPT (Radford et al. 2018)), and encoder-decoder transformers (Vaswani et al. 2017). Many works applied the transformer and resulted in significant translation precision. Overall, results showed that the encoder-decoder transformer architecture (Chen et al. 2022) outperformed all other works.

- *Non-Autoregressive Network:* In 2022, (Huang et al. 2022) implemented a non-autoregressive network to generate sign videos based on glosses, which contains a length regulator and a spatial-temporal graph convolutional generator to produce the sign videos. This model avoids the sequential translation issue faced by transformers by generating all output tokens in parallel, resulting in significantly lower latency during inference (Gu et al. 2018). Although non-autoregressive networks can provide faster results, they may be less accurate when handling complex sentence structures or long-distance dependencies. The trade-off between speed and accuracy is critical when choosing the appropriate model for a given application.

In summary, the applied translation algorithms show that neural machine translation is the most common in the SLMT systems, with the dominance of the transformer architecture. This proves its effectiveness in SLMT due to its ability to capture contextual information. In addition, various performance metrics are implemented to evaluate the models. BLEU-1 and BLEU-4 are the most used, emphasizing that n-gram precision and phrase-level correctness are critical concerns in SLMT. It is worth noting that the bigger the n-gram, the more challenging it is to achieve precision. We also found that PHOENIX-2014T dataset is the most common, as shown in Fig. 10. Therefore, in Fig. 11, we compare the works that tested their models on this dataset. In the T2G translation, the GRU applied in Stoll et al. (2020) slightly outperformed the encoder-decoder transformer fusion applied in Huang et al. (2022). On the other hand, in G2T translation, the encoder-decoder transformer applied in Chen et al. (2022) outperformed the decoder-only transformer applied in Camgoz et al. (2020), the encoder-decoder transformer fusion applied in Huang et al. (2022), and the GRU applied in Camgoz et al. (2018). However, (Huang et al. 2022) reported that the transformer fusion outperformed the original transformer when tested on the same dataset. Moreover, the encoder-decoder transformer in Zhang et al. (2023) outperforms the encoder-only transformer in Camgoz et al. (2020) during S2G translation.

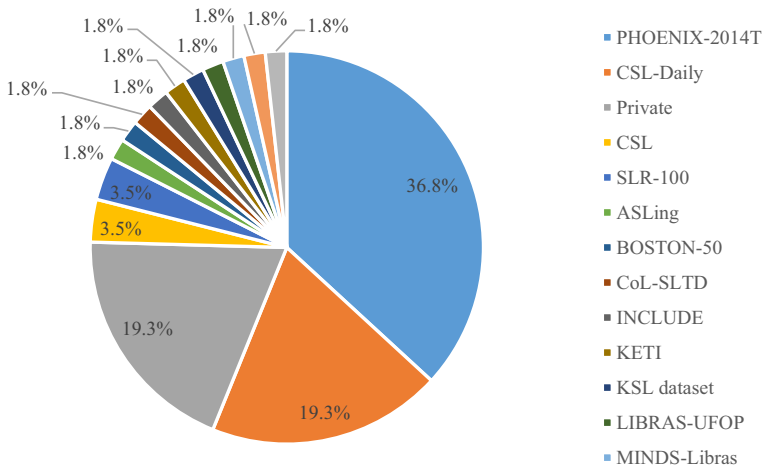


Fig. 10 Datasets frequency in literature

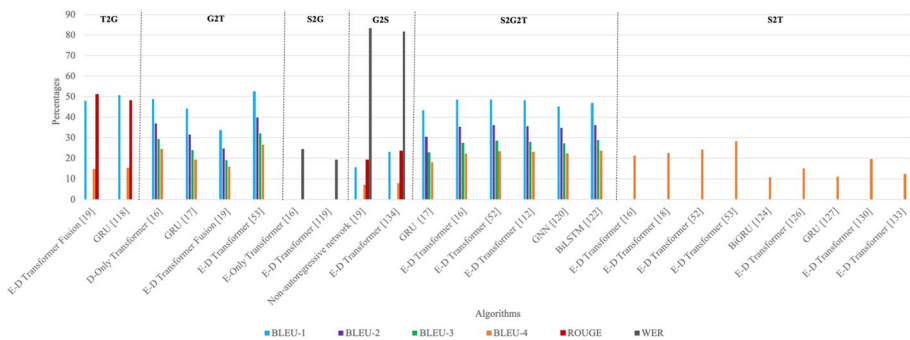


Fig. 11 Comparison between Sign Language Machine Translation Algorithms Based on PHEONIX-2014 T Dataset. E refers to Encoder, and D refers to Decoder

Similarly, in G2S, the encoder-decoder transformer in Tang et al. (2022) outperformed the non-autoregressive network in Huang et al. (2022). The high WER here reflects the difficulty in recognizing context. When comparing the architecture applied in S2G and G2S, we notice that the one in Tang et al. (2022) is the most effective. We believe that this is due to the architecture parameters. On the other hand, in the S2G2T and S2T translations, the encoder-decoder transformers in Zhou et al. (2021) and (Chen et al. 2022) outperform all other architectures and algorithms, respectively. This analysis shows that the encoder-decoder transformer is the most efficient. However, although the transformer in Chen et al. (2022) surpasses all other architectures globally when tested on PHOENIX-2014T dataset, it is not as efficient when tested on CSL-Daily dataset. This is due to the difference in the dataset size, as PHOENIX-2014T dataset (Camgoz et al. 2018) is 3.3 times larger than CSL-Daily dataset (Zhou et al. 2021). Lastly, most works on S2G2T translation (Li and Meng Oct. 2022; Zhou et al. 2021, 2022; Chen et al. 2022; Zhang et al. 2023; Kan et al. 2022) and S2T translation (Li and Meng 2022; Zhou et al. 2021; Chen et al. 2022; Klomsae et al. 2017; Castro et al. 2023; Zhao et al. 2022; Jin et al. 2022; Yin et al. 2022;

Kim and Baek 2023; Chaudhary et al. 2022) consider various sign languages, addressing the need to create multilingual SLMT models.

9 Taxonomy of transformer architectures for sign language machine translation

We present a classification of the component-based Transformer architectures in Fig. 12. We classify these architectures into four categories: (1) Encoder-decoder transformer (EDT) (Fig. 12a), (2) Encoder-only transformer (EOT) (Fig. 12b), (3) Decoder-only transformer (DOT) (Fig. 12c), and (4) Encoder-decoder transformer fusion (EDTF) (Fig. 12d). Table 4 presents a comparison between the works that used these transformer architectures, along with their parameters, in the different stages of our proposed framework.

The EDT architecture was proposed by Vaswani et al. (2017). The architecture consists of a *Tokenization* block to transform the input sentences into words or sub-words that build the model's vocabulary. In addition, the architecture contains an *Embedding* block to transform the input sentences into vectors, plus a *Positional Encoding* block to provide information about the position of each word in the sentence. The model also consists of *Encoder* and *Decoder* layers. The encoder consists of the following components: *Multi-Head Attention* to weigh the importance of different parts of the input and capture the relationships between the input words. *Feedforward Neural Networks* add non-linearities and help the model capture more complex patterns. *Normalization and Residual Connections Layers* to stabilize and speed up the training of deep networks. The decoder consists of components similar to the encoder and a *Masked Multi-Head Attention* to prevent the model from accessing information about future positions. Following the decoder, the model consists of a *Linear Layer* to transform the decoder output into the desired dimension and a *SoftMax* activation function to obtain the probability distribution over the vocabulary for each position in the output. This architecture has been widely used across various translation stages, including S2G (Chen et al. 2022; Zhang et al. 2023), G2T (Chen et al. 2022), G2S (Tang et al. 2022), S2G2T (Li and Meng 2022), and S2T (Li and Meng Oct. 2022; Zhou et al. 2021; Chen et al. 2022; Zhao et al. 2022; Jin et al. 2022; Tang et al. 2022).

The EOT and DOT can be applied in a model that utilizes other algorithms. For instance, the GPT architecture consists of a Byte Per Encoding and a decoder-only transformer (Radford et al. 2018). However, in the SLMT literature, the encoder-only and decoder-only transformers were applied in one work (Camgoz et al. 2020). The researchers here applied the encoder for the S2G translation. The encoder was trained using a CTC loss to predict the gloss sequences. They also applied the decoder for the G2T translation. They trained and tested each part independently and joined them in an end-to-end S2G2T SLMT system.

Furthermore, (Huang et al. 2022) implemented a variation of the encoder-decoder transformer that we call an "encoder-decoder transformer fusion" (EDTF) for G2T, T2G, and G2S translations. The model includes a Fast Fourier Transform (FFT) block between the embedding layer and the encoder. This is added to extract each word's frequency representation (Meesad and Yen 2000) as an additional feature to the embeddings. To enhance the gloss translation, the researchers incorporated a gated bilateral fusion layer within the decoder between the self-attention and the feed-forward network. This mechanism controls the influence of the input and its context while controlling the gradients of the propagation (Li et al. 2019). In addition, they added a CTC joint training layer after the encoder.

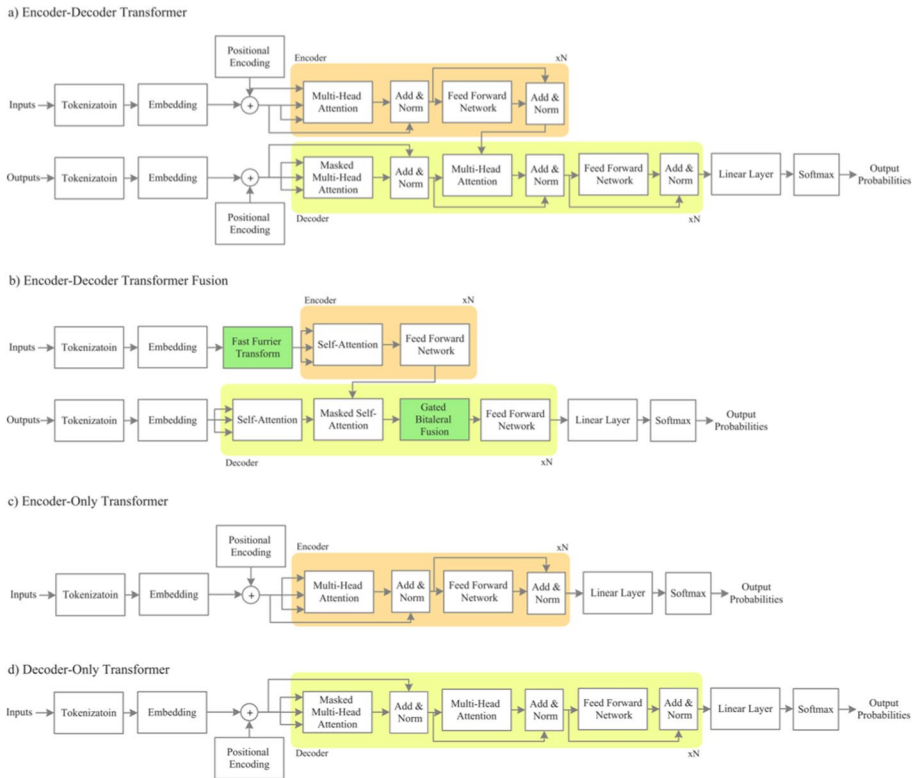


Fig. 12 Taxonomy of Transformer architectures

CTC identifies the repetition of the output and blank outputs and aims to rectify the alignment between the input and output; hence, it allows the model to converge early (Kim et al. 2017). Moreover, they excluded the residual connections and normalization layers in their architecture.

In summary, the Transformer is the most common algorithm in SLMT systems. Overall, the parameters of the encoder-decoder transformer applied in Chen et al. (2022) are the most effective. However, no works compare all four types of transformers in each translation phase i.e. S2G, G2T, T2G, G2S, S2T, and S2G2T translations. This gap should be filled in future works to find the most efficient and accurate SLMT system.

10 Performance evaluation of transformers for gloss-to-text translation: a case study

To understand the implications of deploying transformers-based sign language interpretation in real scenarios, we conduct empirical evaluations of the four transformer architectures which are underpinned by our taxonomy, in a unified environmental setup. It is necessary to deploy an efficient transformer in real-world scenarios. In this case study, we consider sign language gloss to spoken language text (G2T) machine learning translation.

Table 4 Hyperparameters of the Transformer architectures

Work	Algo- rithm	SLMT Covered				Architecture										Dropout	Learning rate	Epochs	Number of itera- tions	Batch Size	Weight decay
		Sign-to- Gloss	Gloss- Text	Text-to- Gloss	Sign- Text	Sign-to- Text	Number of encod- ers	Number of decod- ers	Hidden units	Number of heads	Num- ber of attention blocks	Opti- mizer									
Huang et al. (2022)	EDTF	✗	✓	✓	✗	✗	NR	NR	256	8	2	Adam	NR	0.001	NR	NR	NR	NR			
Chen et al. (2022)	EDT	✓	✓	✗	✗	✓	12	12	1024	16	NR	NR	NR	NR	NR	NR	NR	NR			
Camgoz et al. and (2020)	EOT DOT	✓	✓	✗	✓	✗	NR	NR	512	8	NR	Adam	0.1	10 ⁻⁶	NR	100	32	10 ⁻³			
Zhang et al. (2023)	EDT	✓	✗	✗	✗	✗	2	2	512	8	NR	Adam	0.4	NR	NR	NR	NR	NR			
Li and Meng (2022)	EDT	✗	✗	✗	✓	✓	3	3	512	8	NR	Adam	0.3 and 0.6	10 ⁻⁷	100	NR	32	10 ⁻³			
Zhou et al. (2021)	EDT	✗	✗	✗	✗	✓	NR	NR	512	8	NR	Adam	0.1	5×10 ⁻⁵	30	NR	32	10 ⁻⁶			
Zhao et al. (2022)	EDT	✗	✗	✗	✗	✓	3	6	NR	NR	NR	Adam	NR	3×10 ⁻⁵	300	NR	100	NR			
Jin et al. (2022)	EDT	✗	✗	✗	✗	✓	1	1	512	8	1	Adam	NR	5×10 ⁻⁴	NR	NR	4	NR			

Table 4 (continued)

Work	Algo- rithm	SLMT Covered				Architecture															
		Sign-to- Gloss	Gloss-to- Text	Gloss-to- Text	Sign-to- Gloss	Text-to- Text	Text-to- Text	Sign-to- Text	Sign-to- Text	Number of encod- ers	Number of decod- ers	Hidden units	Number of heads	Num- ber of attention blocks	Opti- mizer	Dropout	Learning rate	Epochs	Number of itera- tions	Batch Size	Weight decay
Chaud- hary et al. (2022)	EDT	✗	✗	✗	✗	✗	✗	✗	✓	3	3	512	8	NR	Adam	0.001	NR	70–150	NR	NR	0.001
	EDT	✗	✗	✗	✓	✗	✗	✗	✗	2	2	512	4	NR	Adam	NR	10 ⁻³	NR	NR	NR	NR

EDT encoder-decoder transformer, EOT encoder-only transformer, DOT decoder-only transformer, EDTF encoder-decoder transformer fusion, NR not reported

10.1 Datasets

We use the largest publicly available sign language PHOENIX-2014T dataset (Camgoz et al. 2018). It consists of 8,257 weather-related sentences in German sign language. To assess the impact of a small dataset on the comparative performance of the different transformer architectures, we employ random 500 unique sentences of PHOENIX-2014T dataset. To evaluate the performance on a different type of dataset and language, we collect our own private ASL dataset, which we call “*MedASL*”. It consists of 500 medical-related sentences generated by ChatGPT (OpenAI 2023) to reflect scenarios between patients and doctors, nurses, technicians, and registration desk staff in a medical center. An ASL expert then translated these sentences to gloss.

10.2 Experimental setup

To achieve the best performance possible in a sign language interpretation scenario, we perform hyperparameter tuning which determines the optimal values for transformer models’ parameters. The parameters we study for each transformer model are presented in Table 5. These parameters are selected as the best parameters reported in the state of the art as described in Table 4. Consequently, for each dataset, we perform 16 experiments, corresponding to each combination of the four architectures and their respective hyperparameters configurations.

10.3 Experimental results analysis

Figure 13 shows our results for the four transformer architectures with the best hyperparameters configuration. EDT architecture has the best performance across all datasets. This reveals that EDT maintains performance consistency regardless of the dataset size. This is because the encoder captures the gloss context then the decoder produces the translated text more efficiently. Such architecture is crucial for sequence-to-sequence tasks like SLMT. In contrast, EDTF has the worst performance. This is due to omitting the positional encoding and normalization layers, which leads to translation inefficiency. This underperformance indicates fundamental limitations in the integration of the encoder and decoder networks or incompatibility of the fusion approach with G2T translation. DOT performed better than EOT across all datasets. This is due to the capability of the decoder to generate sequences token by token based on previous tokens. However, EOT is not suitable for translating sequences and focuses on input understanding and representation.

In summary, for SLMT G2T translation, EDT outperforms all other architectures due to its capability to combine context understanding of gloss and sequence-to-sequence translation. Regarding hyperparameter tuning, our results reveal that EDT and DOT architectures’ performance increased with the increasing number of heads and hidden units.

11 Challenges and proposed solutions

A real-time efficient SLMT system comprises a user-centric architecture that allows DHH and hearing users to interact with each other through a simultaneous translation from sign to spoken language and vice versa. The challenges of such a system include:

Table 5 Hyperparameters in the literature for the Transformer architectures under study

Algorithm	Hyperparameters										Assigned name for hyperparameters combination		
	Number of encoders	Number of decoders	Hidden units	Number of heads	Number of attention blocks	Optimizer	Dropout	Learning rate	Batch size	Weight decay		Hyperparameter tuning	Justification for hyperparameter selection
EDT (Huang et al. 2022)	NR, 1 *	NR, 1 *	256	8	2	Adam	NR, 0 *	0.001	32	NR, 0 *	✗	NR	H1
EDTF (Chen et al. 2022)	12	12	1024	16	NR, 1 *	NR, Adam *	NR, 0 *	0.001	NR, 32 **	NR, 0 *	✗	NR	H2 H3
DOT (Camgoz et al. 2020)	NR, 1 *	NR, 1 *	512	8	NR, 1 *	Adam	0.1	10 ⁻⁶	32	10 ⁻³	✗	NR	H4

EDT encoder-decoder transformer, EDTF encoder-decoder transformer fusion, EOT encoder-only transformer, DOT decoder-only transformer, H1 hyperparameters configuration 1, H2 hyperparameters configuration 2, H3 hyperparameters configuration 3, H4 hyperparameters configuration 4, NR not reported
*Default value as it was not reported in literature. **The same as the optimal value in other architectures

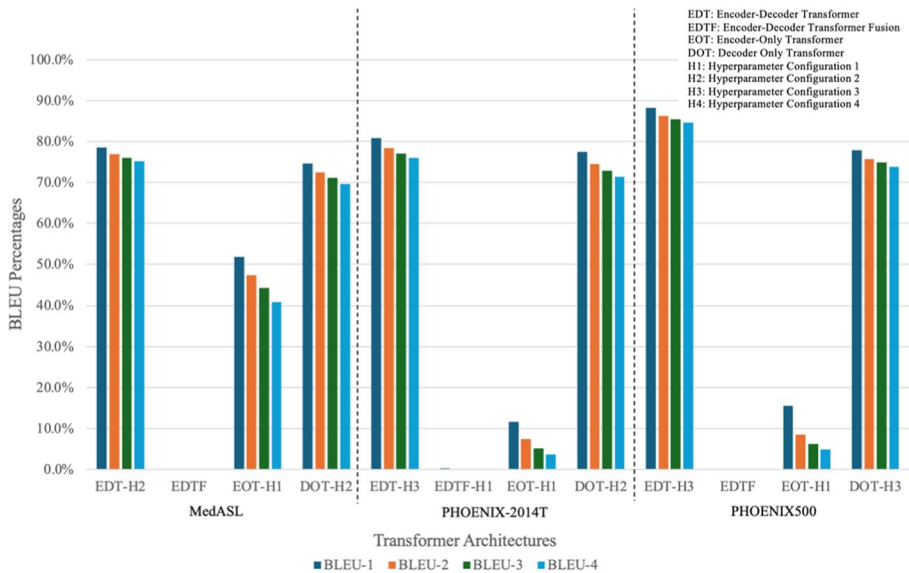


Fig. 13 Comparison between the four transformer architectures for gloss-to-text translation with the best hyperparameter configuration on MedASL, PHOENIX-2014T, and PHOENIX500 datasets

Availability of large and multilingual datasets: There are more than 300 sign languages globally (Kumar et al. 2015), many of which are low in resources, such as the ArSL (“Arabic Sign Language Framework”, Mada Qatar Assistive Technology Center. 2023). Consequently, the number of available datasets is scarce. Therefore, researchers need a large and accurate dataset for faster data generation and precision (Amin et al. 2022). In addition, sign language datasets often include the translation between one spoken language and one sign language (e.g., Chinese and Chinese Sign Language) (Núñez-Marcos et al. 2023). Consequently, limitations of current technologies include the training of monolingual AI models, in which models are usually trained on one pair of languages at a time. A solution for this challenge is the creation of crowdsourced platforms (Sosoni et al. 2018) for collecting, annotating, and translating sign language data. This approach overcomes the scarcity of resources in a particular sign language. It leads to the creation of a large multilingual dataset, which reduces the need for extensive datasets for each language. Another solution is to apply transfer learning techniques by pre-training a model on rich sign language datasets and then fine-tuning them on low-resource sign language (Chen et al. 2022). This strategy is particularly relevant for including regional dialects and idiomatic expressions, which are critical for the accurate representation of sign languages in AI models.

- High deployment cost:** The deployment of SLMT systems involves substantial financial investment, due to the advanced technology required for accurate sign language recognition and translation and the need for extensive infrastructure, including servers for processing and storing large datasets. The economic barrier extends to the cost of developing, testing, and continually updating these systems to accommodate new sign languages and dialects, making it a significant obstacle to widespread adoption. A potential solution to mitigate these costs is utilizing cloud and edge computing to reduce the need for expensive, dedicated infrastructure (Ismail et al. 2008), and by dis-

tributing the computational load more efficiently and cost-effectively across edge and cloud servers (Materwala et al. 2023).

- **High retraining cost:** Languages are constantly evolving (Aitchison 2005). Therefore, the current datasets will soon need to be updated, and current AI models must be retrained. This leads to high electricity and maintenance costs. (Chen et al. 2022) applied transfer learning techniques to grasp the knowledge from rich sign languages and apply it to low-resource languages. Through this approach, the model could learn universal sign language features, including the translation of idiomatic expressions and regional dialects. On the other hand, domain adaptation strategies allow the AI model to adjust to a specific in-domain low-resource sign language and out-of-domain translations for the same sign language with high translation accuracy (Chu and Wang 2018).
- **High energy consumption:** SLMT systems involve Internet-of-Things (IoT) devices and edge and cloud servers to train the AI models and perform real-time translation. These components consume a high amount of energy. By 2025, global data centers will consume around 800 terawatt-hours of electrical energy, triple the consumption in 2021 (International Energy Agency 2022). A solution for such a challenge is the introduction of energy-aware strategies for resource distribution and allocation (Ismail and Abed 2019).
- **QoS and latency management:** An ultra-low latency is essential yet challenging for a real-time simultaneous translation, especially in critical situations. In the case of SLMT, sentences in sign languages have different structures from spoken languages (Jay 2011), making it challenging to start the translation before a sentence ends. (Niehues et al. 2018) proposed a solution to apply reinforcement learning for sequence-level optimization in addition to multi-task learning, which allows a model to be trained on multiple tasks simultaneously, such as POS tagging and machine translation. This solution facilitates domain adaptation and transfer learning, leading to faster training convergence, improved translation quality, and reduced latency. Another solution is to train the dataset on the Cloud while processing the lightweight translation on Edge servers near the translation device (Ismail and Buyya 2023). However, to our knowledge, no work provides a comparative analysis between a human interpreter and SLMT system in terms of execution time, leaving a gap in determining the efficiency of SLMT systems in real-world scenarios. Conducting such a study is necessary to provide a clearer picture of current SLMT system capabilities and areas where they still lag behind human performance.
- **Data privacy and security:** Security mechanisms should be in place for SLMT systems as the translation exposes personal information and surroundings during video transmission (Bragg et al. 2020). Therefore, intelligent approaches are required to automate security responses in the IoT network. Security threats in such scenarios include API attacks, eavesdropping, and DDoS attacks can be mitigated through intrusion detection and digital forensics techniques (Al-Doghman et al. 2023). In gathering sign language data, collecting consent forms from system users is crucial for DHH individual privacy. In a peer-to-peer environment where the systems are used by multiple stakeholders, such as healthcare professionals, in communication with the DHH individual, it is very important to ensure the integrity of the consent forms. A promising approach is the employment of blockchain technology that enables a privacy-preserving and secure environment among peers without intermediate third-party using the consensus protocol, and the reinforcement of consent rules via the execution of blockchain smart contracts (Ismail and Materwala 2020). This method not only provides a secured record of consent but also empowers users with the ability to grant, modify, or

transparently withdraw their consent, thus offering an additional layer of security and trust in the consent process.

- **User acceptance:** Achieving widespread user acceptance SLMT systems presents a significant challenge. A survey by Kacorri et al. (2017) showed that demographics, such as educational background, and technology experiences significantly influence acceptance of automated SLMT systems. Furthermore, it revealed frequent sign language users tend to be more critical of the automated systems due to higher QoS expectations and authenticity of sign language representation. The general skepticism towards automated SLMT systems as replacements for live interpreters or in telephone relay services further complicates the acceptance, as participants are concerned about the system's ability to convey complex sign language nuances accurately and effectively (Kacorri et al. 2015). To overcome this challenge, we propose adopting a participatory design approach that involves DHH users in the development process. This is to ensure that the SLMT systems are tailored to meet the diverse needs and expectations of the DHH community.

12 Future research directions

In this section, we identify several promising directions for future research in this field. A critical area of focus is the development of more robust and scalable end-to-end SLMT systems that can handle a wide range of sign languages efficiently in terms of precision and execution time. While Transformer is the most used approach for sign language translation, as revealed by our survey, Reinforcement learning would aid in leveraging Transformer translation (Chen et al. 2021) for sign language translation. In addition, integrating sign language features, such as facial expressions and body language, could further enhance the accuracy and naturalness of SLMT systems.

Transformer models have shown exceptional ability in understanding and generating complex spoken language patterns (Vaswani et al. 2017), which could be extended to the nuances of sign languages, with the potential of increasing the translation precision of an LSMT system. However, training the Transformer large datasets suffers from performance efficiency in terms of execution time, which hinders its deployment in real-time SLMT systems. Quantum computing is a promising technology to boost the performance of computing power-hungry applications (Oshurko 2020). It could reduce the computational time required for training deep learning models on large datasets due to their processing power. Therefore, the development of end-to-end SLMT systems enabled by quantum computing would revolutionize the domain of sign language translation, toward the deployment of more sophisticated SLMT systems while increasing their precision and execution time. This would encourage the adoption of these systems by DHH and hearing populations, enabling real-time conversation.

However, deploying a real-time conversational end-to-end SLMT system faces privacy and security challenges of conversational data and videos collected for the DHH individuals. Blockchain has shown its potential in privacy-preserving health records management systems (Ismail and Materwala 2020). by creating decentralized ledgers to manage and securely store DHH individual data and consent logs, blockchain technology can provide a robust framework for protecting sensitive information while maintaining DHH individual trust in the SLMT system (Ismail et al. 2021).

Furthermore, to provide the users of SLMT systems with immersive, interactive, and engaging experiences for their users, the integration of scalable and real-time Metaverse (Ismail and Buyya 2023), along with Virtual Reality (VR) (Vaitkevicius et al. 2019), Augmented Reality (AR) (Vinayagamoorthy et al. 2018), wearable devices (Lee and Lee 2018), and educational tools (Abbas et al. 2022) holds promise for revolutionizing how the DHH community interacts with the world.

13 Conclusion

The increasing number of Deaf and Hard of Hearing (DHH) worldwide with limited certified sign language interpreters has led to a need for an efficient, signs-driven, integrated end-to-end automated Sign Language Machine Translation (SLMT) system. Many works on the topic gained attention in recent years. Most of the works in the literature proposed systems based on Neural Machine Translation (NMT) algorithms to achieve accurate translation. Our study identifies different Transformer architectures as the most used and effective algorithms in this domain, with PHEONIX-2014 T dataset being the most widely applied.

Through this study, we provide an in-depth analysis of the evolution and current state of Machine Translation (MT) systems. We also reveal key insights, challenges, and future research directions, addressing gaps left by existing surveys that have typically focused on specific aspects of sign language translations, such as signs to glosses, glosses to spoken language text, text to glosses, or glosses to signs. In addition, we propose an end-to-end SLMT framework covering all translation stages, i.e. from sign to gloss to text and backward. This framework allows SLMT researchers and developers to systematically evaluate translation methodologies and ensures a comprehensive understanding of the influence of each stage on the overall effectiveness of the translation system. To our knowledge, this is the first work that offers a comprehensive retrospective analysis of the evolution of SLMT algorithms and introduces a taxonomy of Transformer architectures tailored for SLMT. In addition, it presents a Transformer-based gloss-to-text translation case study, comparing the performance of different architectures through empirical evaluations.

One of the key lessons learned is that developing accurate and efficient SLMT systems requires a deep understanding of the unique linguistic features of sign languages. Our analysis demonstrates that Transformer-based architectures surpass other MT models, underscoring the necessity for expansive, annotated datasets and diverse evaluation metrics for thorough SLMT assessment. Moreover, developing accurate and efficient SLMT systems requires a deep understanding of these linguistic features, and the application of large-scale annotated datasets and multiple evaluation metrics, such as BLEU and ROUGE, is critical for comprehensive system assessment.

By addressing these gaps and highlighting the temporal progression of algorithms and the distinct functionalities of Transformer architectures, our work aims to significantly advance the field of SLMT, providing valuable insights and resources that will aid researchers and developers in building more accurate, efficient, and inclusive end-to-end translation systems for seamless communication between the DHH community and the broader society.

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Data availability A private dataset MedASL was generated during the current study. For queries, please contact the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

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References

- American Sign Language Dataset. Accessed: Jul. 31, 2023. [Online]. Available: <https://www.bu.edu/asllrp/>
- “Arabic Sign Language Framework,” Mada Qatar Assistive Technology Center. Accessed: Nov. 28, 2023. [Online]. Available: <https://jumla.mada.org.qa>
- Abbas A, Sarfraz S, Tariq U (2022) Pakistan sign language translation tool in educational setting: teachers perspective. *J Enabling Technol* 16(1):38–47. <https://doi.org/10.1108/JET-06-2021-0033>
- Abney SP (1992) “Parsing by chunks”, principle-based parsing: computation and psycholinguistics. Psychology Press, London, pp 257–278
- Adaloglou N et al (2022) A Comprehensive study on deep learning-based methods for sign language recognition. *IEEE Trans Multimedia* 24:1750–1762. <https://doi.org/10.1109/TMM.2021.3070438>
- Aitchison J (2005) Language change, in the routledge companion to semiotics and linguistics. Routledge, London, pp 111–120
- Al-Barham M et al. (2023) RGB arabic alphabets sign language dataset, *arXiv preprint arXiv:2301.11932*
- Al-Doghman F, Moustafa N, Khalil I, Sohrabi N, Tari Z, Zomaya AY (2023) AI-enabled secure microservices in edge computing: opportunities and challenges. *IEEE Trans Serv Comput* 16(2):1485–1504. <https://doi.org/10.1109/TSC.2022.3155447>
- Al-Hammadi M, Muhammad G, Abdul W, Alsulaiman M, Bencherif MA, Mekhtiche MA (2020) Hand gesture recognition for sign language using 3DCNN. *IEEE Access* 8:79491–79509. <https://doi.org/10.1109/ACCESS.2020.2990434>
- Almeida SGM, Rezende TM, Almeida GTB, Toffolo ACR, Guimarães FG (2019) MINDS-libras dataset. Zenodo. <https://doi.org/10.5281/zenodo.2667329>
- Alves V, Ribeiro J, Faria P and Romero L (2022) Neural machine translation approach in automatic translations between portuguese language and portuguese sign language glosses, in 17th Iberian Conference on Information Systems and Technologies (CISTI), IEEE, 1–7.
- Amin MS, Rizvi STH, Hossain MdM (2022) A comparative review on applications of different sensors for sign language recognition. *J Imaging* 8(4):98. <https://doi.org/10.3390/jimaging8040098>
- Ananthanarayana T, Kotecha N, Srivastava P, Chaudhary L, Wilkins N and Nwogu I (2021) Dynamic Cross-Feature Fusion for American Sign Language Translation, in 16th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2021), IEEE, 1–8. <https://doi.org/10.1109/FG52635.2021.9667027>.

- Arivazhagan N et al. (2019) Massively Multilingual Neural Machine Translation in the Wild: Findings and Challenges, arXiv preprint [arXiv:1907.05019](https://arxiv.org/abs/1907.05019)
- Athitsos V et al. (2008) The American Sign Language Lexicon Video Dataset, in 2008 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, IEEE, Jun. 1–8. <https://doi.org/10.1109/CVPRW.2008.4563181>.
- Bahdanau D, Cho K and Bengio Y (2014) Neural machine translation by jointly learning to align and translate, arXiv preprint [arXiv:1409.0473](https://arxiv.org/abs/1409.0473)
- Balas V, Kumar R, Srivastava R (2020) Recent trends and advances in artificial intelligence and internet of things. Springer, Berlin
- Bhagwat SR, Bhavsar RP and Pawar BV (2021) Translation from Simple Marathi sentences to Indian Sign Language Using Phrase-Based Approach, in 2021 International Conference on Emerging Smart Computing and Informatics (ESCI), IEEE, 367–373. <https://doi.org/10.1109/ESCI50559.2021.9396900>.
- Bhuiyan MN, Rahman MM, Billah MM, Saha D (2021) Internet of things (IoT): A review of its enabling technologies in healthcare applications, standards protocols, security, and market opportunities. IEEE Internet Things J 8(13):10474–10498
- Boulares M, Jemni M (2016) Learning sign language machine translation based on elastic net regularization and latent semantic analysis. Artif Intell Rev 46(2):145–166. <https://doi.org/10.1007/s10462-016-9460-3>
- Bragg D, Koller O, Caselli N and Thies W (2020) Exploring collection of sign language datasets: privacy, participation, and model performance, in The 22nd International ACM SIGACCESS Conference on Computers and Accessibility, New York, NY, USA: ACM, 1–14. <https://doi.org/10.1145/3373625.3417024>.
- Brown PF, Della Pietra SA, Della Pietra VJ, Mercer RL (1993) The mathematics of statistical machine translation: parameter estimation. Comput Linguist 19(2):263–311
- Brown T, Mann B, Ryder N, Subbiah M and Kaplan J (2020) Language models are few-shot learners. in 34th Conference on Neural Information Processing Systems
- Camgoz NC, Kindiroglu AA, Karabu'klu S, Kelepir M, O'zsoy AS and Akarun L (2016) BosphorusSign: A Turkish sign language recognition corpus in health and finance domains, in Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), 1383–1388
- Camgoz NC, Hadfield S, Koller O, Ney H and Bowden R (2018) Neural Sign Language Translation, in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 7784–7793.
- Camgoz N, Koller O, Hadfield S and Bowden R (2020) Sign language transformers: joint end-to-end sign language recognition and translation, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10023–10033.
- Carl M, Way A (2003) Recent advances in example-based machine translation. Springer, Dordrecht. <https://doi.org/10.1007/978-94-010-0181-6>
- Cerna LR, Cardenas EE, Miranda DG, Menotti D, Camara-Chavez G (2021) A multimodal LIBRAS-UFOP Brazilian sign language dataset of minimal pairs using a microsoft Kinect sensor. Expert Syst Appl 167:114179. <https://doi.org/10.1016/j.eswa.2020.114179>
- Chai X, Wang H and Chen X (2014) The devisign large vocabulary of chinese sign language database and baseline evaluations
- Charniak E et al (1996) Taggers for parsers. Artif Intell 85:45–57
- Chaudhary L, Ananthanarayana T, Hoq E, Nwogu I (2022) SignNet II: a transformer-based two-way sign language translation model. IEEE Trans Pattern Anal Mach Intell. <https://doi.org/10.1109/TPAMI.2022.3232389>
- Chen Y, Wei F, Sun X, Wu Z and Lin S (2022) A simple multi-modality transfer learning baseline for sign language translation, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 5120–5130.
- Chen L et al. (2021) Decision transformer: Reinforcement learning via sequence modeling, in Advances in neural information processing systems, 15084–15097
- Cho K et al. (2014) Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation, in Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), Stroudsburg, PA, USA: Association for Computational Linguistics, 1724–1734. <https://doi.org/10.3115/v1/D14-1179>.
- Chu C and Wang R (2018) A survey of domain adaptation for neural machine translation, in Proceedings of the 27th International Conference on Computational Linguistics, Association for Computational Linguistics, pp. 1304–1319.
- Cooper H, Ong E-J, Pugeault N, Bowden R (2012) Sign language recognition using sub-units. J Mach Learn Res 13:2205–2231

- Costa REO et al (2019) Towards an open platform for machine translation of spoken languages into sign languages. *Mach Transl* 33(4):315–348. <https://doi.org/10.1007/s10590-019-09238-5>
- Dahmani D, Larabi S (2014) User-independent system for sign language finger spelling recognition. *J Vis Commun Image Represent* 25(5):1240–1250. <https://doi.org/10.1016/j.jvcir.2013.12.019>
- Dankers V, Lucas C, and Titov I (2022) Can transformer be too compositional? Analysing Idiom Processing in Neural Machine Translation,” in Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Stroudsburg, PA, USA: Association for Computational Linguistics, 3608–3626. <https://doi.org/10.18653/v1/2022.acl-long.252>.
- de Castro GZ, Guerra RR, Guimarães FG (2023) Automatic translation of sign language with multi-stream 3D CNN and generation of artificial depth maps. *Expert Syst Appl* 215:119394. <https://doi.org/10.1016/j.eswa.2022.119394>
- Devan P, Khare N (2020) An efficient XGBoost–DNN-based classification model for network intrusion detection system. *Neural Comput Appl* 32(16):12499–12514. <https://doi.org/10.1007/s00521-020-04708-x>
- Dongxu L, Cristian R, Xin Y and Hongdong L (2020) Word-level deep sign language recognition from video: a new large-scale dataset and methods comparison, in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 1459–1469.
- Dreuw P, Deselaers T, Keysers D and Ney H (2006) Modeling image variability in appearance-based gesture recognition, in ECCV workshop on statistical methods in multi-image and video processing, 7–18.
- Dreuw P, Neidle C, Athitsos V, Sclaroff S and Ney H (2008) Benchmark databases for video-based automatic sign language recognition. in *LREC*
- A. Duarte *et al.*, “How2Sign: a large-scale multimodal dataset for continuous American sign language,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, 2021, 2735–2744.
- Elakkiya R, Natarajan B (2021) ISL-CSLTR: Indian sign language dataset for continuous sign language translation and recognition. Mendeley Data. <https://doi.org/10.17632/kcmpdxky7p.1>
- “Ethnologue.” Accessed: Oct. 08, 2023. [Online]. Available: <https://www.ethnologue.com>
- El-Alfy E-SM, Luqman H (2022) A comprehensive survey and taxonomy of sign language research. *Eng Appl Artif Intell* 114:105198. <https://doi.org/10.1016/j.engappai.2022.105198>
- Fang B, Co J and Zhang M (2017) DeepASL: enabling ubiquitous and non-intrusive word and sentence-level sign language translation, in Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems, New York, NY, USA: ACM, 1–13. <https://doi.org/10.1145/3131672.3131693>.
- Farooq U, Rahim MSM, Sabir N, Hussain A, Abid A (2021) Advances in machine translation for sign language: approaches, limitations, and challenges. *Neural Comput Appl* 33(21):14357–14399. <https://doi.org/10.1007/s00521-021-06079-3>
- Farooq U, Mohd Rahim MS, Abid A (2023) A multi-stack RNN-based neural machine translation model for English to Pakistan sign language translation. *Neural Comput Appl* 35(18):13225–13238. <https://doi.org/10.1007/s00521-023-08424-0>
- Gough N and Way A (2004) Robust large- scale EBMT with marker-based segmentation, in Proceedings of the Tenth Conference on Theoretical and Methodological Issues in Machine Translation (TMI-04), 95–104.
- Gu J, Bradbury J, Xiong C, Li V and Socher R (2018) Non-autoregressive neural machine translation, in *ICLR*
- Guo D, Zhou W, Li A, Li H, Wang M (2020) Hierarchical recurrent deep fusion using adaptive clip summarization for sign language translation. *IEEE Trans Image Process* 29:1575–1590. <https://doi.org/10.1109/TIP.2019.2941267>
- Guo D, Zhou W, Li H and Wang M (2018) Hierarchical LSTM for Sign Language Translation, in Proceedings of the AAAI Conference on Artificial Intelligence. <https://doi.org/10.1609/aaai.v32i1.12235>
- Hassan S, Berke L, Vahdani E, Jing L, Tian Y, and Huenerfauth M (2020) An isolated-signing RGBD dataset of 100 American Sign Language signs produced by fluent ASL signers, in *LREC*, 89–94.
- Healthy Hearing, “Hearing loss statistics at a glance,” Healthy Hearing. Accessed: Oct. 27, 2023. [Online]. Available: <https://www.healthyhearing.com/report/52814-Hearing-loss-statistics-at-a-glance>
- Hochreiter S, Schmidhuber J (1997) Long short-term memory. *Neural Comput* 9(8):1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Huang J, Zhou W, Zhang Q, Li H and Li W (2018) Video-based sign language recognition without temporal segmentation, Proceedings of the AAAI Conference on Artificial Intelligence, vol. 32, no. 1, <https://doi.org/10.1609/aaai.v32i1.11903>.
- Huang W, Zhao Z, He J, and Zhang M (2022) DualSign: Semi-Supervised Sign Language Production with Balanced Multi-Modal Multi-Task Dual Transformation, in Proceedings of the 30th ACM

- International Conference on Multimedia, New York, USA: ACM, 5486–5495. <https://doi.org/10.1145/3503161.3547957>.
- Ibrahim NB, Selim MM, Zayed HH (2018) An automatic arabic sign language recognition system (ARSLRS). *J King Saud Univ* 30(4):470–477. <https://doi.org/10.1016/j.jksuci.2017.09.007>
- International Energy Agency, “Data Centres and Data Transmission Networks,” 2022. Accessed: May 29, 2023. [Online]. Available: <https://www.iea.org/reports/data-centres-and-data-transmission-networks>
- Ismail L, Abed EH (2019) Linear power modeling for cloud data centers: taxonomy, locally corrected linear regression, simulation framework and evaluation. *IEEE Access* 7:175003–175019. <https://doi.org/10.1109/ACCESS.2019.2956881>
- Ismail L, Materwala H (2020) Blockchain paradigm for healthcare: performance evaluation. *Symmetry (basel)* 12(8):1200. <https://doi.org/10.3390/sym12081200>
- Ismail L, Zhang L (2018) Information innovation technology in smart cities. Springer, Singapore. <https://doi.org/10.1007/978-981-10-1741-4>
- Ismail L, Materwala H, Karduck AP, Adem A (2020) Requirements of health data management systems for biomedical care and research: scoping review. *J Med Internet Res*. <https://doi.org/10.2196/17508>
- Ismail L, Materwala H, Hennebelle A (2021) A scoping review of integrated blockchain-cloud (BcC) architecture for healthcare: applications, challenges and solutions. *Sensors* 21(11):3753
- Ismail L and Buyya R (2023) Metaverse: a vision, architectural elements, and future directions for scalable and realtime virtual worlds, *ArXiv*
- Ismail L, Mills B and Hennebelle A (2008) A formal model of dynamic resource allocation in Grid computing environment,” in Proceedings of the 2008 9th ACIS International Conference on Software Engineering, Artificial Intelligence, Networking, and Parallel/Distributed Computing
- Ismail L, Masud MM and Khan L (2014) FSBD: a framework for scheduling of big data mining in cloud computing, Proceedings - 2014 IEEE International Congress on Big Data, BigData Congress 2014, 514–521, <https://doi.org/10.1109/BigData.Congress.2014.81>.
- Jay M (2011) Don't Just Sign... Communicate!: a student's guide to mastering ASL grammar. Judea Media, Burbank
- Jin T, Zhao Z, Zhang M and Zeng X (2022) MC-SLT: towards low-resource signer-adaptive sign language translation, in Proceedings of the 30th ACM International Conference on Multimedia, New York, NY, USA: ACM, pp. 4939–4947. <https://doi.org/10.1145/3503161.3548069>.
- Johnson M et al (2017) Google's multilingual neural machine translation system: enabling zero-shot translation. *Trans Assoc Comput Linguist* 5:339–351. https://doi.org/10.1162/tacl_a_00065
- Johnston T (2010) From archive to corpus transcription and annotation in the creation of signed language corpora. *Int J Corpus Linguist* 15(1):106–131. <https://doi.org/10.1075/ijcl.15.1.05joh>
- Joze HRV, Koller O (2018) MS-ASL: a large-scale data set and benchmark for understanding American sign language. *ArXiv*. <https://doi.org/10.48550/arXiv.1812.01053>
- Kacorri H, Huenerfauth M, Ebling S, Patel K, Menzies K, Willard M (2017) Regression analysis of demographic and technology-experience factors influencing acceptance of sign language animation. *ACM Trans Access Comput* 10(1):1–33. <https://doi.org/10.1145/3046787>
- Kacorri H, Huenerfauth M, Ebling S, Patel K and Willard M (2015) Demographic and experiential factors influencing acceptance of sign language animation by deaf users, in Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility - ASSETS '15, New York, New York, USA: ACM Press, 147–154. <https://doi.org/10.1145/2700648.2809860>.
- Kahlon NK, Singh W (2023) Machine translation from text to sign language: a systematic review. *Univers Access Inf Soc* 22(1):1–35. <https://doi.org/10.1007/s10209-021-00823-1>
- Kamata K, Yoshida T, Watanabe M, and Usui Y (1989) An approach to Japanese-sign language translation system, in Conference Proceedings., IEEE International Conference on Systems, Man and Cybernetics, IEEE, 1089–1090. <https://doi.org/10.1109/ICSMC.1989.71466>.
- Kan J, Hu K, Hagenbuchner M, Tsoi AC, Bennamoun M and Wang Z (2022) Sign language translation with hierarchical spatio-temporal graph neural network, in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 3367–3367.
- Kapitanov A, Kvanchiani K, Nagaev A and Petrova E (2023) Slovo: Russian Sign Language Dataset, arXiv preprint [arXiv:2305.14527](https://arxiv.org/abs/2305.14527)
- Katwe PK, Khamparia A, Gupta D, Dutta AK (2023) Methodical systematic review of abstractive summarization and natural language processing models for biomedical health informatics: approaches, metrics and challenges. *ACM Trans Asian Low-Resour Lang Inform Proc*. <https://doi.org/10.1145/3600230>
- Kay M (1984) Functional unification grammar: A formalism for machine translation, in 10th International Conference on Computational Linguistics and 22nd Annual Meeting of the Association for Computational Linguistics, 75–78.

- Khan RU, Khattak H, Wong WS, AlSalman H, Mosleh MAA, Md. Mizanur Rahman SK (2021) Intelligent malaysian sign language translation system using convolutional-based attention module with residual network. *Comput Intell Neurosci* 2021:1–12. <https://doi.org/10.1155/2021/9023010>
- Kim Y, Baek H (2023) Preprocessing for keypoint-based sign language translation without glosses. *Sensors* 23(6):3231. <https://doi.org/10.3390/s23063231>
- Kim S, Hori T and Watanabe S (2017) Joint CTC-attention based end-to-end speech recognition using multi-task learning. in IEEE international conference on acoustics, speech and signal processing (ICASSP), 4835–4839.
- Klakow D, Peters J (2002) Testing the correlation of word error rate and perplexity. *Speech Commun* 38(1–2):19–28. [https://doi.org/10.1016/S0167-6393\(01\)00041-3](https://doi.org/10.1016/S0167-6393(01)00041-3)
- Klomsae A, Auephanwiriyaikul S, Theera-Umpon N (2017) A novel string grammar unsupervised possibilistic C-medians algorithm for sign language translation systems. *Symmetry* 9(12):321. <https://doi.org/10.3390/sym9120321>
- Ko S-K, Kim CJ, Jung H, Cho C (2019) Neural sign language translation based on human keypoint estimation. *Appl Sci* 9(13):2683. <https://doi.org/10.3390/app9132683>
- Koehn P (2009) Statistical machine translation. Cambridge University Press, Cambridge
- Koller O, Forster J, Ney H (2015) Continuous sign language recognition: Towards large vocabulary statistical recognition systems handling multiple signers. *Comput vis Image Underst* 141:108–125. <https://doi.org/10.1016/j.cviu.2015.09.013>
- Kouremenos D, Ntalianis K, Kollias S (2018) A novel rule based machine translation scheme from Greek to Greek sign language: production of different types of large corpora and language models evaluation. *Comput Speech Lang* 51:110–135. <https://doi.org/10.1016/j.csl.2018.04.001>
- Kumar VK, Goudar RH, Desai VT (2015) Sign language unification: the need for next generation deaf education. *Procedia Comput Sci* 48:673–678. <https://doi.org/10.1016/j.procs.2015.04.151>
- Kurakin A, Zhang Z and Liu Z (2012) A real time system for dynamic hand gesture recognition with a depth sensor, in Proceedings of the 20th European Signal Processing Conference (EUSIPCO) 1975–1979.
- Lane H, Howard C and Max Hapke H (2018) Natural Language Processing in Action Understanding, analyzing, and generating text with Python, [Online]. Available: www.manning.com
- Lass R (1984) Phonology: an introduction to basic concepts. Cambridge University Press, Cambridge
- Latif G, Mohammad N, Alghazo J, AlKhalaf R, AlKhalaf R (2019) ArASL: Arabic alphabets sign language dataset. *Data Brief* 23:103777. <https://doi.org/10.1016/j.dib.2019.103777>
- Lee BG, Lee SM (2018) Smart wearable hand device for sign language interpretation system with sensors fusion. *IEEE Sens J* 18(3):1224–1232. <https://doi.org/10.1109/JSEN.2017.2779466>
- Li G, Zhu L, Liu P and Yang Y (2019) Entangled transformer for image captioning, in Proceedings of the IEEE/Cvf International Conference on Computer Vision, 8928–8937.
- Li J, Liu Y, Xu W and Li Z (2022) GASLA: Enhancing the Applicability of Sign Language Translation, in IEEE INFOCOM 2022 - IEEE Conference on Computer Communications, IEEE, 1249–1258. <https://doi.org/10.1109/INFOCOM48880.2022.9796819>
- Li R, Meng L (2022) Sign language recognition and translation network based on multi-view data. *Appl Intell* 52(13):14624–14638. <https://doi.org/10.1007/s10489-022-03407-5>
- Lin C-Y (2004) Rouge: A package for automatic evaluation of summaries, in Text summarization branches out, 74–81.
- Lopez A (2008) Statistical machine translation. *ACM Comput Surv* 40(3):1–49. <https://doi.org/10.1145/1380584.1380586>
- Luqman H, Mahmoud SA (2019) Automatic translation of Arabic text-to-arabic sign language. *Univers Access Inf Soc* 18(4):939–951. <https://doi.org/10.1007/s10209-018-0622-8>
- Materwala H, Ismail L, Hassanein HS (2023) QoS-SLA-aware adaptive genetic algorithm for multi-request offloading in integrated edge-cloud computing in internet of vehicles. *Veh Commun* 43:1–21. <https://doi.org/10.1016/j.vehcom.2023.100654>
- Matthes S et al. (2012) Dicta-Sign -Building a Multilingual Sign Language Corpus, in 5th Workshop on the Representation and Processing of Sign Languages: Interactions between Corpus and Lexicon. Satellite Workshop to the eighth International Conference on Language Resources and Evaluation
- Mavi A (2020) A New Dataset and Proposed Convolutional Neural Network Architecture for Classification of American Sign Language Digits, *ArXiv*, 2020.
- Mavi A and Dikle Z (2022) A New 27 Class Sign Language Dataset Collected from 173 Individuals, *ArXiv*
- Meesad P, Yen GG (2000) Pattern classification by a neurofuzzy network: application to vibration monitoring. *ISA Trans* 39(3):293–308. [https://doi.org/10.1016/S0019-0578\(00\)00027-6](https://doi.org/10.1016/S0019-0578(00)00027-6)
- Morrissey S and Way A (2005) An example-based approach to translating sign language, in Workshop on example-based machine translation, 109–116.

- Natarajan B, Elakkiya R (2022) Dynamic GAN for high-quality sign language video generation from skeletal poses using generative adversarial networks. *Soft Comput* 26(23):13153–13175. <https://doi.org/10.1007/s00500-022-07014-x>
- National Research Council (1966) *Language and machines: computers in translation and linguistics*. National Academies Press, Washington
- Neco RP and Forcada ML (1997) Asynchronous translations with recurrent neural nets, in *Proceedings of International Conference on Neural Networks (ICNN'97)*, IEEE, 2535–2540. <https://doi.org/10.1109/ICNN.1997.614693>.
- Niehues J, Pham NQ, Ha T-L, Sperber M and Waibel A (2018) Low-Latency Neural Speech Translation, in *Interspeech 2018, ISCA: ISCA*, 1293–1297. <https://doi.org/10.21437/Interspeech.2018-1055>.
- Nirenburg S (1989) Knowledge-based machine translation. *Mach Transl* 4(1):5–24. <https://doi.org/10.1007/BF00367750>
- Nunes Vieira L, O'Sullivan C, Zhang X, O'Hagan M (2023) Privacy and everyday users of machine translation. *Trans Spaces* 12(1):21–44. <https://doi.org/10.1075/ts.22012.nun>
- Núñez-Marcos A, Perez-de-Viñaspre O, Labaka G (2023) A survey on Sign Language machine translation. *Expert Syst Appl* 213:118993. <https://doi.org/10.1016/j.eswa.2022.118993>
- Oliveira M et al. (2017) A Dataset for Irish sign language recognition, in *Irish Machine Vision and Image Processing Conference (IMVIP)*
- OpenAI (2023) GPT-4 Technical Report
- OpenAI, “ChatGPT.” Accessed: May 31, 2023. [Online]. Available: <https://chat.openai.com>
- Oshurko I (2020) Quantum machine learning, in *Quantum Computing*, 209–234. https://doi.org/10.1142/9781786348210_0010.
- Oszust M and Wysocki M (2013) Polish sign language words recognition with Kinect, in *6th International Conference on Human System Interactions (HSI)*, pp. 219–226.
- Othman A and Jemni M (2012) English-asl gloss parallel corpus 2012: Aslg-pc12, in *LREC*, European Language Resources Association (ELRA), 151–154.
- Papastratis I, Chatzikonstantinou C, Konstantinidis D, Dimitropoulos K, Daras P (2021) Artificial intelligence technologies for sign language. *Sensors* 21(17):5843. <https://doi.org/10.3390/s21175843>
- Papineni K, Roukos S, Ward T and Zhu W-J (2001) BLEU: a method for automatic evaluation of machine translation, in *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics - ACL '02*, Morristown, NJ, USA: Association for Computational Linguistics, 311. <https://doi.org/10.3115/1073083.1073135>.
- Parker JR (2010) *Algorithms for image processing and computer vision*. Wiley, Hoboken
- Poibeau T (2017) *Machine translation*. MIT Press, Cambridge
- Pu J, Zhou W and Li H (2016) Sign Language Recognition with Multi-modal Features, 252–261. https://doi.org/10.1007/978-3-319-48896-7_25.
- Pugeault N and Bowden R (2011) Spelling it out: Real-time ASL fingerspelling recognition, in *2011 IEEE International Conference on Computer Vision Workshops (ICCV Workshops)*, IEEE, 1114–1119. <https://doi.org/10.1109/ICCVW.2011.6130290>.
- Radford A, Wu J, Child R, Luan D (2019) Language models are unsupervised multitask learners. *OpenAI Blog* 1:9
- Radford A, Narasimhan K, Salimans T and Sutskever I (2018) Improving language understanding by generative pre-training
- Rastgou R, Kiani K, Escalera S (2020) Hand sign language recognition using multi-view hand skeleton. *Expert Syst Appl* 150:113336. <https://doi.org/10.1016/j.eswa.2020.113336>
- Registry of Interpreters for the Deaf, “Registry,” Registry of Interpreters for the Deaf. Accessed: Oct. 27, 2023. [Online]. Available: <https://rid.org>
- Rodriguez J et al. (2020) Understanding motion in sign language: a new structured translation dataset, in *Proceedings of the Asian Conference on Computer Vision (ACCV)*
- Sandler W, Lillo-Martin DC (2006) *Sign language and linguistic universals*. Cambridge University Press, Cambridge
- Scarselli F, Gori M, Chungtsui Ah, Hagenbuchner M, Monfardini G (2009) The graph neural network model. *IEEE Trans Neural Netw* 20(1):61–80. <https://doi.org/10.1109/TNN.2008.2005605>
- Schembri A, Fenlon J, Rentelis R, Reynolds S, Cormier K (2013) *Building the British sign language corpus*. University of Hawaii Press, Honolulu
- Shahin N and Ismail L (2023) ChatGPT, Let us chat sign language: experiments, architectural elements, challenges and research Directions,” in *2023 International Symposium on Networks, Computers and Communications (ISNCC)*, IEEE, 1–7. <https://doi.org/10.1109/ISNCC58260.2023.10323974>.

- Sheridan P, Smeaton AF (1992) The application of morpho-syntactic language processing to effective phrase matching. *Inf Process Manag* 28(3):349–369. [https://doi.org/10.1016/0306-4573\(92\)90080-J](https://doi.org/10.1016/0306-4573(92)90080-J)
- Sidig AAI, Luqman H, Mahmoud S, Mohandes M (2021) KArSL: Arabic sign language database. *ACM Trans Asian Low-Resour Lang Inform Proc* 20(1):1–19. <https://doi.org/10.1145/3423420>
- Sincan OM, Keles HY (2020) AUTSL: a large scale multi-modal turkish sign language dataset and baseline methods. *IEEE Access* 8:181340–181355. <https://doi.org/10.1109/ACCESS.2020.3028072>
- Sosoni V et al. (2018) Translation crowdsourcing: creating a multilingual corpus of online educational content, in *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*
- Sridhar A, Ganesan RG, Kumar P and Khapra M (2020) INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition, in *Proceedings of the 28th ACM International Conference on Multimedia*, New York, NY, USA: ACM, 1366–1375. <https://doi.org/10.1145/3394171.3413528>.
- Stahlberg F (2020) Neural machine translation: a review. *J Artif Intell Res* 69:343–418. <https://doi.org/10.1613/jair.1.12007>
- Stoll S, Camgoz NC, Hadfield S, Bowden R (2020) Text2Sign: towards sign language production using neural machine translation and generative adversarial networks. *Int J Comput vis* 128(4):891–908. <https://doi.org/10.1007/s11263-019-01281-2>
- Sutskever I, Vinyals O, Le QV (2014) Sequence to sequence learning with neural networks. *Adv Neural Inf Process Syst*. <https://doi.org/10.48550/arXiv.2301.08912>
- Tabak J (2014) *Geometry: the language of space and form*. Infobase Publishing, New York
- Tang S, Hong R, Guo D and Wang M. (2022) Gloss semantic-enhanced network with online back-translation for sign language production, in *Proceedings of the 30th ACM International Conference on Multimedia*, New York, NY, USA: ACM, 5630–5638. <https://doi.org/10.1145/3503161.3547830>.
- Thangali A, Nash JP, Sclaroff S and Neidle C (2011) Exploiting phonological constraints for handshape inference in ASL video, in *CVPR 2011*, IEEE, 521–528. <https://doi.org/10.1109/CVPR.2011.5995718>.
- Tonkin E (2017) The importance of medical interpreters. *Am J Psychiatry Resid J* 12(8):13–13. <https://doi.org/10.1176/appi.ajp-rj.2017.120806>
- Vaitkevičius A, Taroza M, Blažauskas T, Damaševičius R, Maskeliūnas R, Woźniak M (2019) Recognition of american sign language gestures in a virtual reality using leap motion. *Appl Sci* 9(3):445. <https://doi.org/10.3390/app9030445>
- Vaswani A, Shazeer N, Parmar N, Uszkoreit J (2017) Attention is All You Need. *Adv Neural Inform Proc Syst*. <https://doi.org/10.48550/arXiv.1706.03762>
- Vicars W (2023) American Sign Language: ‘car’ / ‘drive,’” *Lifepoint.com*. Accessed: Aug. 03, 2023. [Online]. Available: <https://www.lifepoint.com/asl101/pages-signs/c/cardrive.htm>
- Vicars W (2023) ASL: Classifiers. Accessed: Aug. 03, 2023. [Online]. Available: <https://www.lifepoint.com/asl101/pages-signs/classifiers/classifiers-main.htm>
- Viitaniemi V, Jantunen T, Savolainen L, Karppa M and Laaksonen J, S-pot: a benchmark in spotting signs within continuous signing, in *LREC, European Language Resources Association*
- Vinayagamoorthy V, Glancy M, Debenham P, Bruce A, Ziegler C and Schäffer R (2018) Personalising the TV experience with augmented reality technology, in *Proceedings of the 2018 ACM International Conference on Interactive Experiences for TV and Online Video*, New York, NY, USA: ACM, 179–184. <https://doi.org/10.1145/3210825.3213562>.
- Wang H, Wu H, He Z, Huang L, Church KW (2022) Progress in Machine Translation. *Engineering* 18:143–153. <https://doi.org/10.1016/j.eng.2021.03.023>
- Wolk K, Marasek K (2015) Neural-based machine translation for medical text domain. Based on european medicines agency leaflet texts. *Procedia Comput Sci* 64:2–9. <https://doi.org/10.1016/j.procs.2015.08.456>
- World Health Organization, “Deafness and hearing loss,” World Health Organization. Accessed: Oct. 27, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss>
- Yang S, Jung S, Kang H, Kim C (2020) The Korean sign language dataset for action recognition. Springer International Publishing, Cham, pp 532–542. https://doi.org/10.1007/978-3-030-37731-1_43
- Yin A, Zhao Z, Jin W, Zhang M, Zeng X and He X (2022) MLSLT: towards multilingual sign language translation, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 5109–5119
- Yin A et al. (2021) SimulSLT: end-to-end simultaneous sign language translation, in *Proceedings of the 29th ACM International Conference on Multimedia*, New York, NY, USA: ACM, 4118–4127. <https://doi.org/10.1145/3474085.3475544>.

- Zadeh LA (1965) Fuzzy sets. *Inf Control* 8(3):338–353
- Zahedi M, Keysers D, Deselaers T and Ney H (2005) Combination of tangent distance and an image distortion model for appearance-based sign language recognition, in *Pattern Recognition: 27th DAGM Symposium, Vienna, Austria, August 31-September 2, 2005. Proceedings 27*, Springer, 401–408
- Zens R, Och FJ, and Ney H (2002) Phrase-Based Statistical Machine Translation, in *KI 2002: Advances in Artificial Intelligence: 25th Annual German Conference on AI, KI 2002 Aachen, Germany, September 16–20, 2002 Proceedings 25*, Springer, 18–32. https://doi.org/10.1007/3-540-45751-8_2.
- Zhang Z et al (2023) Universal multimodal representation for language understanding. *IEEE Trans Pattern Anal Mach Intell*. <https://doi.org/10.1109/TPAMI.2023.3234170>
- Zhang J, Zhou W, Xie C, Pu J and Li H (2016) Chinese sign language recognition with adaptive HMM, in 2016 IEEE International Conference on Multimedia and Expo (ICME), IEEE, 1–6. <https://doi.org/10.1109/ICME.2016.7552950>.
- Zhang B, Müller M and Sennrich R (2023) SLTUNET: a simple unified model for sign language translation, in *International Conference on Learning Representations*
- Zhao J, Qi W, Zhou W, Duan N, Zhou M, Li H (2022) Conditional sentence generation and cross-modal reranking for sign language translation. *IEEE Trans Multimedia* 24:2662–2672. <https://doi.org/10.1109/TMM.2021.3087006>
- Zheng J et al (2020) An improved sign language translation model with explainable adaptations for processing long sign sentences. *Comput Intell Neurosci* 2020:1–11. <https://doi.org/10.1155/2020/8816125>
- Zheng J, Chen Y, Wu C, Shi X, Kamal SM (2021) Enhancing neural sign language translation by highlighting the facial expression information. *Neurocomputing* 464:462–472. <https://doi.org/10.1016/j.neucom.2021.08.079>
- Zhou H, Zhou W, Zhou Y, Li H (2022) Spatial-temporal multi-cue network for sign language recognition and translation. *IEEE Trans Multimedia* 24:768–779. <https://doi.org/10.1109/TMM.2021.3059098>
- Zhou H, Zhou W, Qi W, Pu J and Li H (2021) Improving sign language translation with monolingual data by sign back-translation, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 1316–1325.
- Zou H, Hastie T (2005) Regularization and variable selection via the elastic Net. *J R Stat Soc Series B Stat Methodol* 67(2):301–320. <https://doi.org/10.1111/j.1467-9868.2005.00503.x>

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